

EyeCon- The Speaker for Paralyzed

Submitted in partial fulfillment of the requirements of the degree of

BACHELOR OF COMPUTER ENGINEERING

by

Ms. Aditi Godbole IEN:11712007

Ms. Deepika Rao IEN: 11712006

Ms. Komal Pawoolkar IEN: 11712035

Guide:

Dr. Sanjay Sharma



Department of Computer Engineering

New Horizon Institute of Technology and Management,

Thane (2020-2021)



NEW HORIZON INSTITUTE OF TECHNOLOGY AND MANAGEMENT, THANE

CERTIFICATE

This is to certify that the project entitled “EyeCon- The Speaker for Paralyzed” is a bonafide work of **Ms. Aditi Godbole (IEN- 11712007), Ms. Deepika Rao (IEN- 11712006), Ms. Komal Pawoolkar (IEN- 11712035)** submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering in Computer Engineering**.

Dr. Sanjay Sharma

(Guide)

Dr. Sanjay Sharma

(Head of Department)

Dr. Prashant Deshmukh

(Principal)



NEW HORIZON INSTITUTE OF
TECHNOLOGY AND MANAGEMENT,
THANE

Project Report Approval for B.E.

This project report entitled *EyeCon- The Speaker for Paralyzed* by *Ms. Aditi Godbole, Ms. Deepika Rao & Ms. Komal Pawoolkar* is approved for the degree of *Bachelor of Engineering in Computer Engineering, 2020-21.*

Examiner Name:

Signature:

1. _____

2. _____

Date:

Place: Thane

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.



Ms. Aditi Godbole (IEN- 11712007)



Ms. Deepika Rao (IEN- 11712006)



Ms. Komal Pawoolkar (IEN 11712035)

Date: 7 May 2021

Abstract

The fundamental goal of this project is to allow impaired and handicapped patients to convert their intents which will be gauged using eye blinks and converted to text and speech. A plethora of people across the globe are affected by motor function disabilities. Some people also lose their ability to communicate due to some trauma or accidents. This makes communication for paralyzed patients next to impossible. However, people with paralysis can use their eyes as a means of communication. The natural motion of an eye is the eye blink and it can be used efficiently to communicate information. There are various tools which can be used to detect these eye- blinks. Most of the devices these days come with built-in webcams/ front cameras. Our project aims to detect the eye blinks of the patient using webcams or mobile front cameras and convert them to text and speech respectively. The interface detects the voluntary eyeblinks and interprets them as commands which are used to select text for the user. This will be simultaneously converted to speech. We intend to deploy this application on play store and create a cross-platform website for the same. Hence, our system intends to provide users with efficient and effective methods of communication without relying on other individuals and without using heavy hardware.

Abbreviation Notation & Nomenclature

Sr No.	Word	Description
1.	MND	Motor Neuron Disease
2.	LIS	Locked- in Syndrome
3.	ALS	Amyotrophic lateral sclerosis
4.	MS	Multiple Sclerosis
5.	EAR	Eye Aspect Ratio
6.	AAC	Augmentative and Alternative Communication
7.	SRS	Software Requirement Specification

CONTENTS

1	Introduction.....	1
2	Literature Survey.....	3
	2.1 Literature Background.....	3
	2.2 Existing Technologies.....	7
3	Problem Statement, Objective and Scope.....	10
	3.1 Problem Statement & Objective.....	10
	3.2 Scope.....	11
4	Requirement Analysis and Methodology.....	13
	4.1 Requirement Specification.....	13
	4.2 Analysis and Design.....	16
	4.3 Pseudo Codes.....	19
5	Project Planning and Scheduling.....	20
	5.1 Project Planning and Scheduling.....	20
6	System Design and Implementation.....	25
	6.1 System Design.....	25
	6.2 Implementation.....	31
7	Results and Discussion.....	45

7.1 Results.....	44
7.2 Test Cases.....	48
7.3 Discussion.....	52
8 Conclusion.....	54
Future Scope.....	56
References.....	58
Plagiarism Test Report.....	60
Publication by Candidate.....	87
Acknowledgement.....	88

LIST OF FIGURES:

1. Figure 2.1- MindScribe Architecture.....	8
2. Figure 2.2- Descriptive Image for Eye Control.....	8
3. Figure 4.1- Architecture Overview.....	16
4. Figure 4.2- Detailed Architecture	17
5. Figure 4.3- Eye Blink Detection Module.....	17
6. Figure 4.4- User Interaction Module.....	18
7. Figure 4.5-Text to Conversion Module	18
8. Figure 5.2- Graphical Representation of Project Timeline (Aug 2020- Nov 2020)	21
9. Figure 5.4- Graphical Representation of Project Timeline (Jan 2021- April 2021).....	22
10. Figure 6.1- Flow graph of Blink Detection Module.....	26
11. Figure 6.2- Flow graph of User Interaction Module.....	27
12. Figure 6.3 – Data Flow Diagram.....	28
13. Figure 6.4 – Activity Diagram.....	28
14. Figure 6.5 –Class Diagram.....	29
15. Figure 6.6 – Use Case Diagram.....	30
16. Figure 6.7 – Workflow of Blink Detection Module.....	30
17. Figure 6.8 – Workflow of User Interaction Module.....	31
18. Figure 6.9, 6.10, 6.11– Installation of Anaconda.....	31
19. Figure 6.12,6.13 – Installation of OpenCV.....	33
20. Figure 6.14 - Installation of Dlib.....	34
21. Figure 6.15, 6.16 – Calculation of EAR Coordinates.....	34
22. Figure 6.17 – Function EAR.....	36
23. Figure 6.18 – Loading Face Detector.....	36
24. Figure 6.19 – Capturing Eye Blinks.....	36

25. Figure 6.20 – Function for Virtual Keyboard.....	37
26. Figure 6.21 – Function for Special Keywords.....	38
27. Figure 6.22 – Function to calculate row intervals.....	38
28. Figure 6.23 – Function for Text to Speech Conversion.....	39
29. Figure 6.24 – Website Window.....	40
30. Figure 6.25 –Website Window.....	40
31. Figure 6.26 – Launch the Panel.....	41
32. Figure 6.27 – User Instructions.....	41
33. Figure 6.28 – Contact Page.....	42
34. Figure 6.29 –Linking with the website with flask.....	42
35. Figure 6.30 – User Interface.....	43
36. Figure 7.1 – Results of Blink Detection Module.....	45
37. Figure 7.2, 7.3 – Results of User Interaction Module.....	46
38. Figure 7.4, 7.5 - Results of Text to speech conversion Module.....	47

LIST OF TABLES:

1. Table 5.1- Tabular Representation of Project Timeline (Aug 2020- Nov 2020).....	21
2. Table 5.3- Tabular Representation of Project Timeline (Jan 2021- April 2021).....	23
3. Table 7.1 - Tabular Representation of Eye Blink Detection Test cases.....	48
4. Table 7.2 - Tabular Representation of User Interaction Test cases.....	49
5. Table 7.3 - Tabular Representation of Text to Speech Conversion Test cases...	51
6. Table 7.4 - Tabular Representation of Functional Testing Test cases.....	52

Chapter 1

Introduction

Users usually rely on various devices such as a keyboard, mouse etc. to communicate with a system. These devices make the system easily accessible and hence even novice users don't find these systems daunting. However, people suffering from motor disabilities or impairment cannot use these devices as they are not customized keeping their needs in mind. Assistive technology systems of every variety have been proposed and even commercialized to allow people with disabilities to use their limited voluntary motions to communicate with family and friends, access computers, and control TVs and air conditioners, etc.^[1] Hence, an alternative solution is needed to facilitate glitch less functioning for the impaired. Various researches are being conducted across the globe to overcome this situation. While undertaking the development of such devices, it is critical to consider various aspects such as avoiding the use of heavy and bulky instruments, minimizing the use of manual control, avoiding specialized instruments, featuring real-time performance etc. Existing devices like keyboard, mouse, trackball, touchpad and touchscreens require manual control and hence the inability in movement makes it difficult to be used by patients.^[2] Previous methods for eye blink detection estimate eye state as either open or closed or track eye closure events.

Paralyzed people cannot control muscle function in one or more muscle groups. This condition is usually caused by diseases like ALS, multiple sclerosis etc. or even stroke. The Locked-In Syndrome ie. LIS is a form of paralysis in which the patients lose control of nearly all voluntary muscles. Apart from eye blinks and eye movement, they are unable to control any other part of their body which renders them inefficient to communicate. However, the patients having LIS are still cognitively aware i.e. thoughts and ideas are forming in their brain. These patients heavily rely on eye blinks for communication purposes. Usually, the nurses or caretakers are trained and well versed in interpreting and decoding their blinks. However, they cannot be understood by laymen and hence in the absence of nurses or caretakers, LIS patients have no means of self- expression.

In order to facilitate a means of expression for LIS patients, we propose a robust system which will facilitate the patients to convert their expression to text and speech. This project will work by taking the eye blinks of the patients as input through webcam and front cameras of mobile phones and would effectively convert them to the text which would be displayed at the bottom of the screen. This text will be converted to audio which will provide ease of accessibility and ensure that the patient is heard out at all times. This project primarily focuses to provide a means of communication to people of all classes without having to worry about financial aspects or cumbersome hardware.

Chapter 2

Literature Survey

2.1 Literature Background

2.1.1 Mukherjee, K., & Chatterjee, D. (2015, January). Augmentative and alternative communication device based on eye-blink detection and conversion to morse-code to aid paralyzed individuals. In *2015 International Conference on Communication, Information & Computing Technology (ICCICT)* (pp. 1-5). IEEE.^[3]

Speech disorders can result in a vast number of medical ailments like brain damage, paralysis strokes and many other diseases. Various diseases such as Locked-In Syndrome (LIS) and Amyotrophic Lateral Sclerosis (ALS) often lead to disability in speech. The patient

loses the ability to communicate although the brain is not affected. So, they can still think intellectually like a normal human. This paper published from NIT Durgapur specifically focuses on some of the Augmentative and Alternative Communication Devices which is a cost effective solution for the people. This system converts Intents of Eye blinks to a language named Morse Code which is universally known and accepted.

2.1.2 Goyal, K., Agarwal, K., & Kumar, R. (2017, April). Face detection and tracking: Using OpenCV. In *2017 International conference of Electronics,*

Communication and Aerospace Technology (ICECA) (Vol. 1, pp. 474-478). IEEE.^[4]

Face Detection and Recognition is a well-talked topic in the market of AI. Almost in all aspects of the society face recognition plays an important role. This technology has proven to be a lifesaver in the fields of security and surveillance. The above paper's purpose is a detailed study of Face Recognition using OpenCV library. A diagrammatic comparison is performed on various algorithms like Adaboost, Haar Cascades etc. This paper will work as a basic guide for a novice who wants to study about face recognition.

2.1.3 Sharifi, M., Fathy, M., & Mahmoudi, M. T. (2002, April). A classified and comparative study of edge detection algorithms. In *Proceedings. International conference on information technology: coding and computing* (pp. 117-120). IEEE.^[5]

Edge detection is one of the most commonly used operations in image analysis. An edge is defined by a discontinuity in grey level values. This paper introduces a new classification of most important and commonly used edge detection algorithms, namely ISEF, Canny, Marr- Hildreth, Sobel, Kirsch, Lapla1 and Lapla2. These algorithms are classified into five categories and seven types of analysis are performed with raw data of 30 images.

2.1.4 Abu-Faraj, Z. O., Sleiman, H. C. B., Al Katergi, W. M., Heneine,

J. L. D., & Mashaalany, M. J. (2008). A rehabilitative eye-tracking based brain *Communication and Aerospace Technology (ICECA)* (Vol. 1, pp. 474-478). IEEE.^[4]

Face Detection and Recognition is a well-talked topic in the market of AI. Almost in all aspects of the society face recognition plays an important role. This technology has proven to be a lifesaver in the fields of security and surveillance. The above paper's purpose is a detailed study of Face Recognition using OpenCV library. A diagrammatic comparison is performed on various algorithms like Adaboost, Haar Cascades etc. This paper will work as a basic guide for a novice who wants to study about face recognition.

2.1.5 Sharifi, M., Fathy, M., & Mahmoudi, M. T. (2002, April). A classified and comparative study of edge detection algorithms. In *Proceedings. International conference on information technology: coding and computing* (pp. 117-120). IEEE.^[5]

Edge detection is one of the most commonly used operations in image analysis. An edge is defined by a discontinuity in grey level values. This paper introduces a new classification of most important and commonly used edge detection algorithms, namely ISEF, Canny, Marr- Hildreth, Sobel, Kirsch, Lapla1 and Lapla2. These algorithms are classified into five categories and seven types of analysis are performed with raw data of 30 images.

2.1.6 Abu-Faraj, Z. O., Sleiman, H. C. B., Al Katergi, W. M., Heneine, J. L. D., & Mashaalany, M. J. (2008). A rehabilitative eye-tracking based brain-computer interface for the completely locked-in patient. In *Encyclopedia of Healthcare Information Systems* (pp. 1153-1160). IGI Global.^[6]

A LIS affected patient suffers from problems which include the inability to control the voluntary muscles, speech impairment apart from those needed for eye movements and blinks. This paper describes a low-cost eye tracking based brain-

computer interface system for patients affected with LIS.

2.1.7 KARKI, M. A., TIMILSINA, M. A., REDDY, M. V. I. K., HEGDE, M. S., & GUIDE, M. (2018). CONVERTING INTENT TO TEXT USING REAL- TIME EYEBLINK DETECTION.^[7]

This project has been taken up to allow various physically handicapped patients to convert their intent to text using their eye-blink. Many people around the world are affected by motor function disabilities. To use a computing system motor capabilities are a must. Moreover many people cannot communicate as they have met with accidents or have been under some trauma. It is nearly impossible for a paralyzed person to communicate with others. However, people under paralysis can use their eyes. A natural motion of the eye is an eye-blink. This eye- blink can be efficiently used in order to communicate information.

2.1.8 Mohammed, A., & Anwer, S. (2014). Efficient Eye Blink Detection Method for the disabled-helping domain. *Inter. Jour. of Adv. In: Computer Science and Applications*.^[8]

In this paper, we present a real-time method based on some video and image processing algorithms for eye blink detection. The motivation of this research is the need for disabling who cannot control the calls with human mobile interaction directly without the need for hands. A Haar Cascade Classifier is applied for face and eye detection for getting an eye and facial axis information. In addition, the same classifier is used based on Haar- like features to find out the relationship between the eyes and the facial axis for positioning the eyes. An efficient eye tracking method is proposed which uses the position of detected face. Finally, an eye blinking detection

based on eyelids state (close or open) is used for controlling android mobile phones. The method is used with and without smoothing filters to show the improvement of detection accuracy.

2.1.9 A REVIEW ON SMART EYE BLINK SOLUTION FOR MND PATIENT USING PYTHON", *International Journal of Science & Engineering Development Research (www.ijedr.org)*, ISSN:2455-2631,

Vol.4, Issue 3, page no.361 - 364, March-2019, Tambe Samreen Mohammed , Rajeshwari P.^[9]

The growth of technology in the medical field diminishes the difficulties of patients to an immense extent. The disease is named Motor Neuron Disease (MND). One of the major categories of physical disability leading to paralysis. MND patients are unable to do work like talk, walk, express feelings and communicate due to the weakening of muscles. The patient has control only upon his eye blinks, The problems faced by MND patients is obtaining a solution day by day. The broad review on the literature of different solutions for MND patients is described in this paper. First it contains an overview of the project and the techniques used to solve the patient's problem. Limitation and description of algorithms is compared to our researched technique. The smart solution is obtained using python programming with open cv.

2.1.10 Namratha, K. S., Jeevitha, B. S., Sneha, P. H., & Veda, D. Eye blink to voice format communication for paralyzed patients.^[10]

The wide growth of technology in the medication field reduces the difficulties of patients to a large extent. Motor Neuron Disease is one such major class of physical disabilities resulting in dysfunction. MND patients are unable to figure like walk and communicate caused by weakness of muscles. The patient has management solely upon his Eye movement. This technique contains strategies like face detection, eye detection, eye pursuit, conversion of blink to voice, Video

Oculography methodology is employed to create communication between patient and caretaker. The image process module incorporates digital camera and therefore the eye movement- image is captured with OpenCv to get the coordinate of eye ball. The system permits the patients to speak with caretaker victimisation blink pattern converted into voice format The sensible resolution for image process is obtained be Python programming with OpenCV.

2.2: Existing Technologies

2.2.1 : MindScribe

MindScribe is a system which interprets brain waves specifically designed for people

suffering from Locked-In Syndrome (LIS). This device provides three different modes for communication. The first one is expression mode to identify and share emotional state. The second is a yes/no response mode and third is the typing mode. A company named Neurosky has developed this. MindScribe simply has a single headset with sensors that touch the forehead, measuring brain activity.



Fig 2.1 – Mindscribe Architecture

2.2.2 Eye Control

The EyeControl is the first wearable, screenless, assistive communication device. AI-powered eye-tracking technology enables remote communication between ventilated ICU patients and medical staff via the nurse station. A head-mounted infrared camera tracks the eye movements and sends the information to a small processing unit which translates the movements into communication. The bone conduction component provides audio feedback to the user before the communication is transmitted to the output speaker or connected Bluetooth device, all without the need for a screen.



Fig 2.2- Descriptive Image for Eye Control

2.2.2 Shortcomings Of Existing Technologies

- ☐ Many of the existing technologies employ the use of specialized devices.
- ☐ These devices are generally not portable.
- ☐ Sometimes these devices can cause discomfort.
- ☐ Many of the topics researched have used features like Haar cascades to detect faces, which do not offer real time eyeblink detection.

Chapter 3

Problem Statement, Objective and Scope

3.1 Problem Statement and Objective

3.1.1 Problem Statement:

Creating a website for paralyzed patients which converts the intent of eye blinks to text and speech.

3.1.2 Objective:

- **To facilitate impaired people with a means to communicate independently:**

Many paralyzed patients already use eye blinks as a means of communication which can be usually interpreted and decoded by caretakers and nurses. The ALS association even offers a communication guide that relies on eye blinks. With our project, we intend to automate this task. Our software would read the patient's eye blinks and converts them into text and speech respectively. This would allow patients to express their thoughts and ideas with absolute independence. This kind of independence will provide to be morally uplifting and patients who were earlier reticent to communicate might start communicating more often.

- **To provide a financially accessible solution:**

Various companies are coming up with software solutions to track eye movements and blinks and to interpret and decode them. These software devices are usually powered by expensive hardware. Though these devices undoubtedly facilitate LIS victims to communicate, they can be significantly costly and hence inaccessible to people with limited financial means. We intend to overcome that by developing pocket- friendly equipment. The only required peripheral is a basic webcam. Hence, this software intends to cater to immobilized patients of all classes.

3.2 Scope

- ❑ The project captures live stream video using webcam of the patient and effectively captures the eye blinks of the patient.
- ❑ The eye blinks captured are converted to the text which would be displayed at the bottom of the screen. This text is converted to speech to facilitate transferring crucial information to a person who isn't present on the patient's immediate side.
- ❑ A few assumptions have been made that include, the user's face and eyes being completely present in front of the camera at all times and the user's face being away from the screen at a particular set distance at all times in order to detect and

identify eye blinks. These assumptions render the system incompetent to detect eye blinks if the eyes of the user are partially visible or not at a proper set distance.

Chapter 4

Requirement Analysis and Methodology

4.1 Requirement Specification

4.1.1 Software Requirement Specification

A software requirements specification (SRS) is a description of a software system to be developed. It lays out functional and nonfunctional requirements, and may include a set of use cases that describe user interactions that the software must provide. ^[7]A software requirements specification (SRS) is a comprehensive description of the intended purpose and environment for software under development. Software requirements specification establishes the basis for an agreement between customers and contractors or suppliers (in market-driven projects, these roles may be played by the marketing and development divisions) on what the software product is to do as well as what it is not expected to do.^[7]

4.1.1.1 Functional Requirements

Functional requirements define the basic system behaviour. Functional requirements are the product features. They generally include the input calculations, data manipulation and processing. For our system functional requirements are: -

- ❖ Setting up the keyboard on start
- ❖ Resetting the column and row intervals after each selection
- ❖ Detecting blinks accurately
- ❖ Selecting text accurately

4.1.1.2 User Requirements

- ❖ The user shall use a virtual keyboard that shows the rows and columns with animation.
- ❖ The user will have a view of the selected text.

4.1.1.3 Non-Functional Requirements

Nonfunctional requirements specify the quality attributes of a software. They judge the system based on different characteristics.

- 1) **Reliability** - The system should be reliable and true to its content. It should provide accurate mapping of brain signals into proper robotic actions.
- 2) **Robust** - The system should be resistant enough to all sorts of natural as well as physical damage, which in turn leads to less cost and maintenance.
- 3) **Portable** - The device should be small enough to be carried easily anywhere.
- 4) **Cost** - This is the most important function. System should be cost efficient compared to other alternatives available in the market.
- 5) **Minimalistic Design** - The system interface and design must be user friendly to make it more efficient to use.
- 6) **Usability** - System should be easy to use. Minimal training must be provided.
- 7) **Safety** - Since the device is designed for the differently-abled, the system must be free from all parts which are injurious to the body.
- 8) **Availability** - The system should have good connectivity to the internet so

that all the interpreted data can be stored in the cloud.

4.1.1.4 Hardware Requirements

Camera - This is the main component of the system. Without the access to the camera the system won't work. The camera can be a normal web camera attached to a PC or laptop. It can also be the front camera of a smartphone. The camera quality should be above average quality for better understanding.

4.1.1.5 Software Requirements

Anaconda- Anaconda is a free and open software used for performing machine learning programs efficiently. It provides a good package and module management facility which is better for a novice learner.

Flask- Flask is a web application framework written in python. Flask is based on Werkzeug and Jinja2 template engine. This is quite easy to establish a connectivity with the website's front end and back end.

OpenCV- OpenCV (Open Source Computer Vision Library) is a free software for academic and commercial use. It supports various programming languages like C++, Python and Java and is compatible to work on almost all operating systems like Windows, Linux, Mac etc. OpenCV is used by the camera to capture images and convert them into suitable formats.

Dlib- Dlib is a C++ toolkit consisting of different modules suitable for machine learning projects. Generally, Dlib is used for real time applications.

JavaScript, HTML, CSS3, Bootstrap & Ajax - JavaScript & HTML are programming languages used to build dynamic web applications. Ajax is a tool used to connect to the server without reloading the page. We have used CSS3 and Bootstrap to style the HTML document

Python3 -Python is an interpreted, high-level general-purpose programming language. For our implementation it was used for capturing the eye blinks.

4.2 Analysis and Design

4.2.1 Architecture Overview

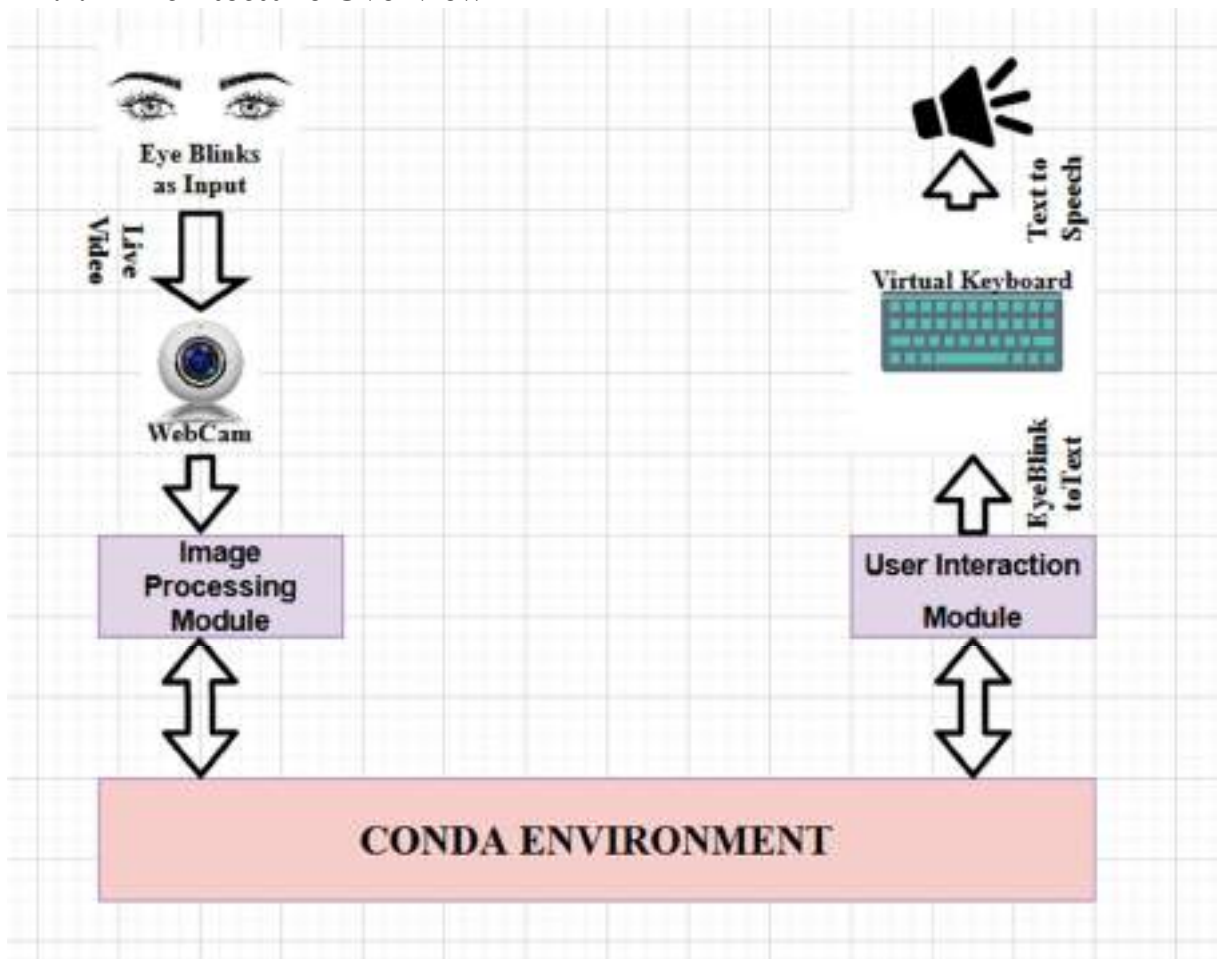


Fig 4.1- Architecture Overview

The vital goal of this project is to take eye blinks of the affected person as input and to transform them to textual content and speech. The same has been depicted by making use of the architecture diagram as shown. The affected person's eye blinks would be examined as input via the Webcam that is then passed onto the image processing module. Then, with the help of the user interaction module, those eye blinks are transformed to textual content after which ultimately to speech.

4.2.2 Detailed Architecture - This provides an overview about how the

system communicates with the client and server.

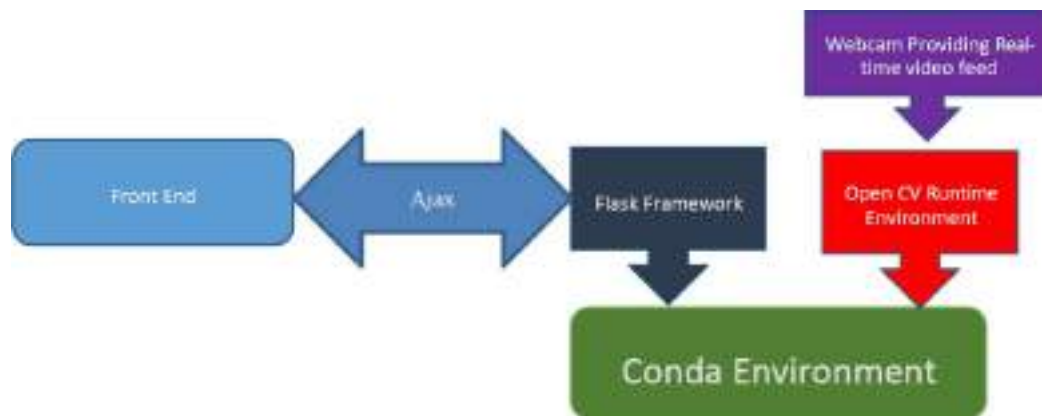


Fig 4.2- Detailed Architecture

4.2.3 Eye Blink Detection Module

The live video captured by the camera is sent to the blink detection module. This processes the video by extracting the facial landmarks using OpenCV, Dlib and other machine learning libraries.

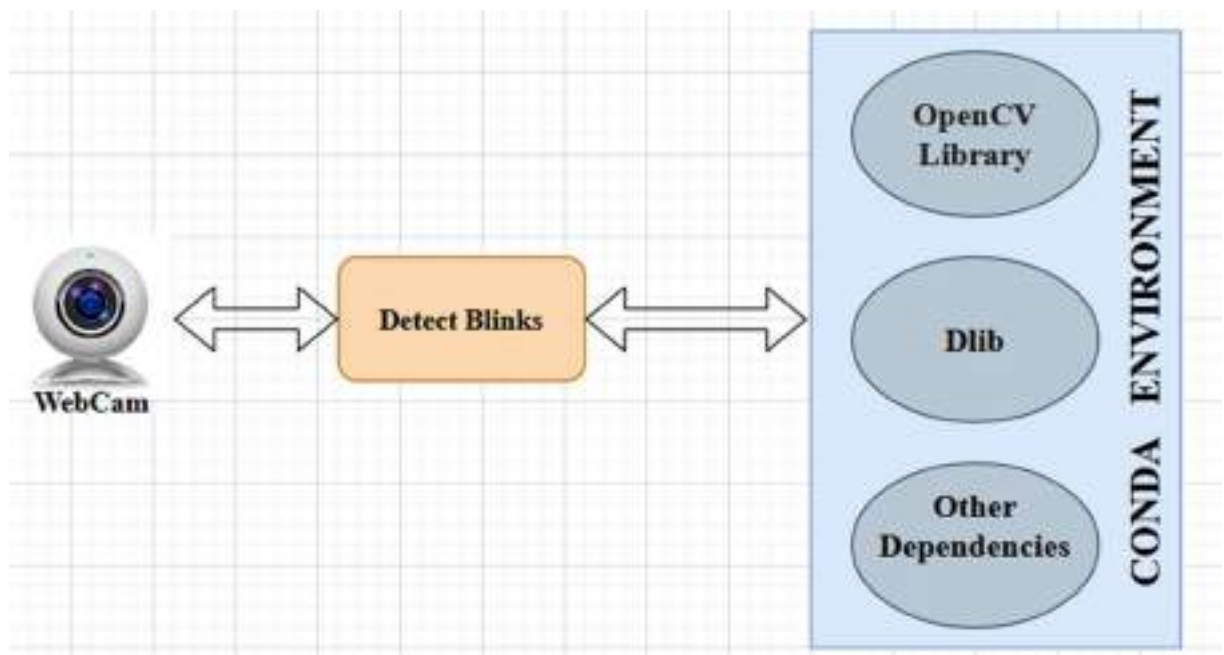


Fig 4.3- Eye Blink Detection Module

4.2.4 User Interaction Module - This provides an overview about the Conda and Flask environment for building the application.

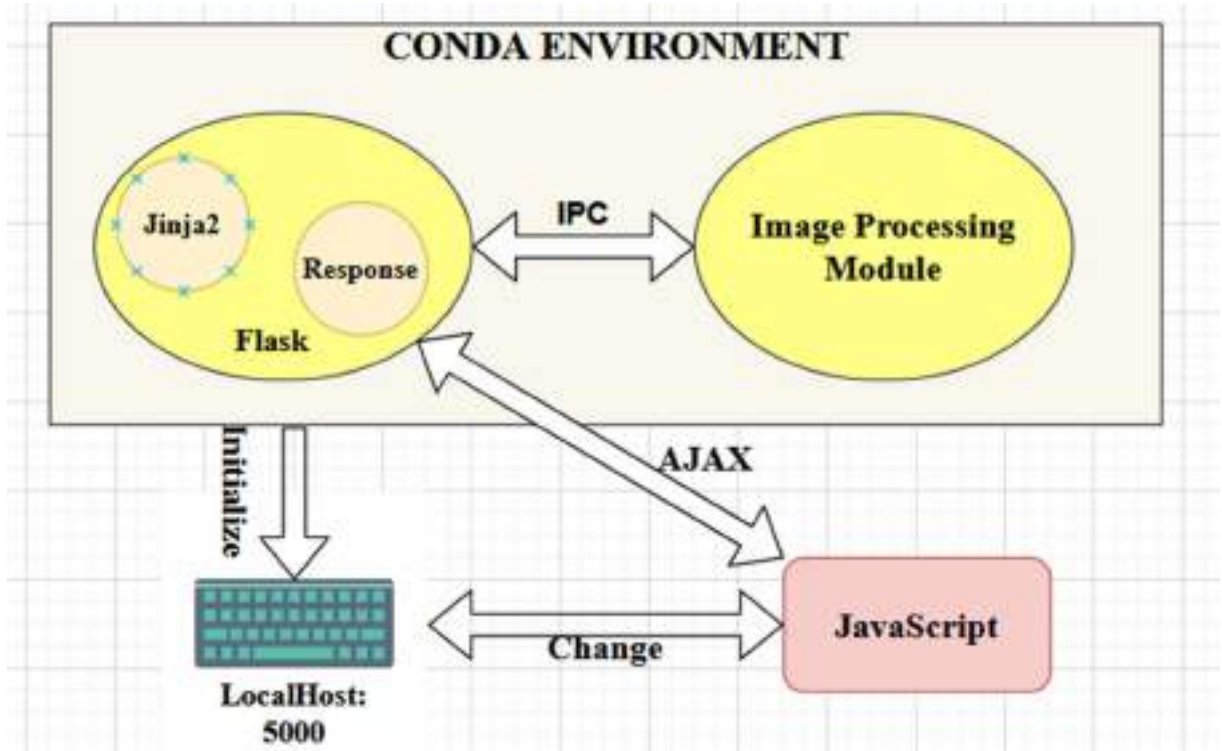


Fig 4.4- User Interaction Module

4.2.5 Text to Speech Conversion Module

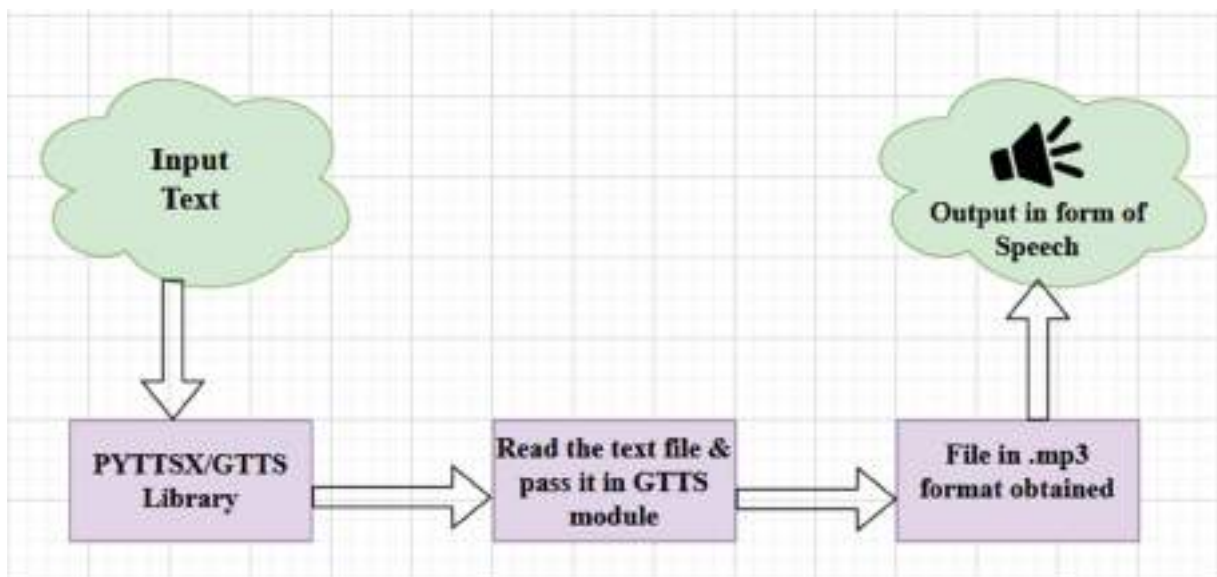


Fig 4.5- Text to Speech Conversion Module

4.3 Pseudo Codes

4.3.1 Eyeblink Detection Module

1. Set EAR_CONSEC_FRAMES & EAR_THRESHOLD
2. While true
3. Grab The frame Resize and grayscale
4. Loop across the faces as the come in the frames
5. Calculate the EAR
6. Create the visualizations across the face
7. If $EAR > EAR_THRESHOLD$
 - a. Increment Counter
8. If $Count > EAR_CONSEC_FRAMES$
 - a. Write into the file

4.3.2 User Interaction Module

1. Initialize keyboard and screen
2. If blink Detected then Select particular column
3. Set column Interval
4. If blink is detected again then select row element
5. Add selected element to the selected text

4.3.3 Text to Speech Conversion Module

1. Once the text is printed on the screen, click on the speech button.
2. If it is a coherent sentence then complete output would be produced as it is.
3. If it is a mixture of random words then the output would be produced individually.

Chapter 5

Project Planning and Scheduling

5.1 Project Planning and Scheduling

This module introduces us to the roadmap of our project. It gives the detailing of tasks undertaken. The project work began in the month of August 2020 and would go on up to April 2021. So far, we have completed the phase 1 of the project which includes Literature Survey, Understanding the pitfalls of the existing technologies and defining our project scope, objective, timeline and the fundamental modules in our project. Phase 2 will include implementation of these modules and deployment of the project. The following tables and Gantt chart would provide an in-depth understanding of the tasks undertaken.

5.1.1 Project Timeline (August 2020- November 2020)

Table 5.1- Tabular Representation of Project Timeline (Aug 2020- Nov 2020)

Title	T/M	Start	End	⌚	%	🔍
PP1 for project topic Selection	T -	03/08/2020	21/08/2020	15 days	100	
Project Approval Report	T -	21/08/2020	04/09/2020	11 days	100	
Understanding literature survey	T -	07/09/2020	11/09/2020	5 days	100	
Scope Redefining and Project Timeline generation	T -	14/09/2020	18/09/2020	5 days	100	
Synopsis Chapter 1 & 2 Preparation.	T -	21/09/2020	25/09/2020	5 days	100	
Project Presentation 1	T -	24/09/2020	25/09/2020	2 days	100	
Study about System Requirements.	T -	28/09/2020	01/10/2020	4 days	100	
Synopsis Chapter 1 & 2 Submission	T -	29/09/2020	05/10/2020	5 days	100	
Study about various algorithms and SDLC	T -	06/10/2020	19/10/2020	10 days	100	
Understanding various technologies	T -	19/10/2020	27/10/2020	7 days	100	
Weekly report submission	T -	28/10/2020	06/11/2020	8 days	100	
Preparation of Synopsis Chapter 3,4 & 5 and Final Draft.	T -	04/11/2020	22/11/2020	13 days	100	
Term End Presentation	T -	23/11/2020	26/11/2020	4 days	100	
Synopsis soft copy generation and submission	T -	27/11/2020	30/11/2020	2 days	100	

Fig 5.1- Graphical Representation of Project Timeline (Aug 2020- Nov 2020)

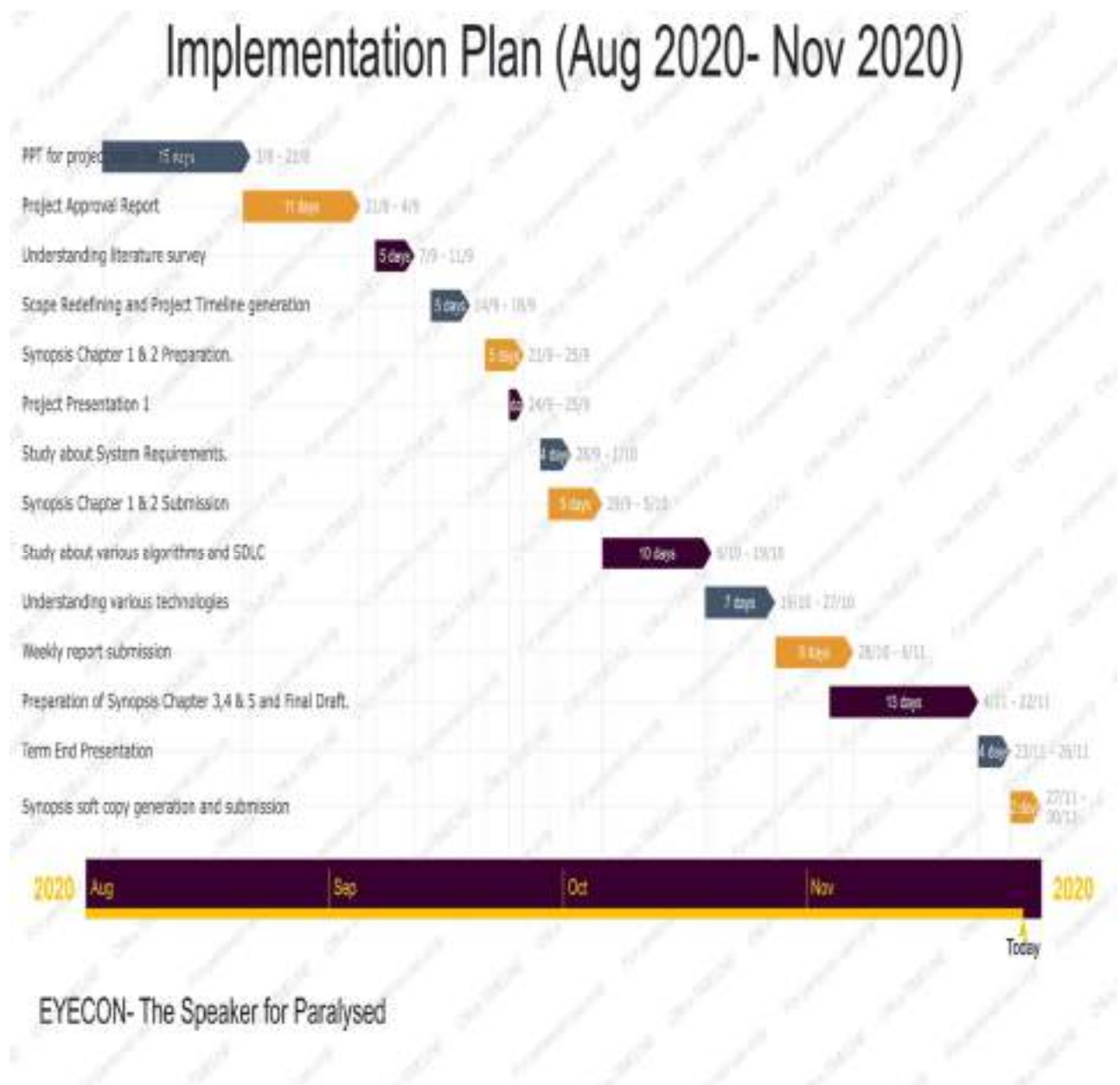


Fig 5.2- Graphical Representation of Project Timeline (Aug 2020- Nov 2020)

5.1.2 Project Timeline (January 2021- April 2021)

Table 5.3- Tabular Representation of Project Timeline (Jan 2021- April 2021)

 Title	T/M	Start	End		%	
 Implementation of Eyeblink Detection Module	 T +	16/01/2021	23/01/2021	5 days	%	
 Implementation of User Interaction Module	 T +	23/01/2021	30/01/2021	5 days	%	
 Connectivity of Front-End & Back-End	 T +	31/01/2021	06/02/2021	5 days	%	
 Conversion from Text to Speech	 T +	07/02/2021	13/02/2021	5 days	%	
 Designing of android application	 T +	12/02/2021	20/02/2021	6 days	%	
 Overall Integration of modules and implementation	 T +	21/02/2021	28/02/2021	5 days	%	
 Testing & Deployment of Applications	 T +	01/03/2021	13/03/2021	10 days	%	
 Completion of Report	 T +	14/03/2021	27/03/2021	10 days	%	
 Working on Research Paper	 T +	28/03/2021	10/04/2021	10 days	%	
 Final Reviewing	 T +	11/04/2021	16/04/2021	5 days	%	
 Project Submission	 T +	19/04/2021	19/04/2021	1 day	%	
 Research Paper Submission	 T +	22/04/2021	23/04/2021	2 days	%	

Fig 5.3- Graphical Representation of Project Timeline (Aug 2020- Nov 2020)

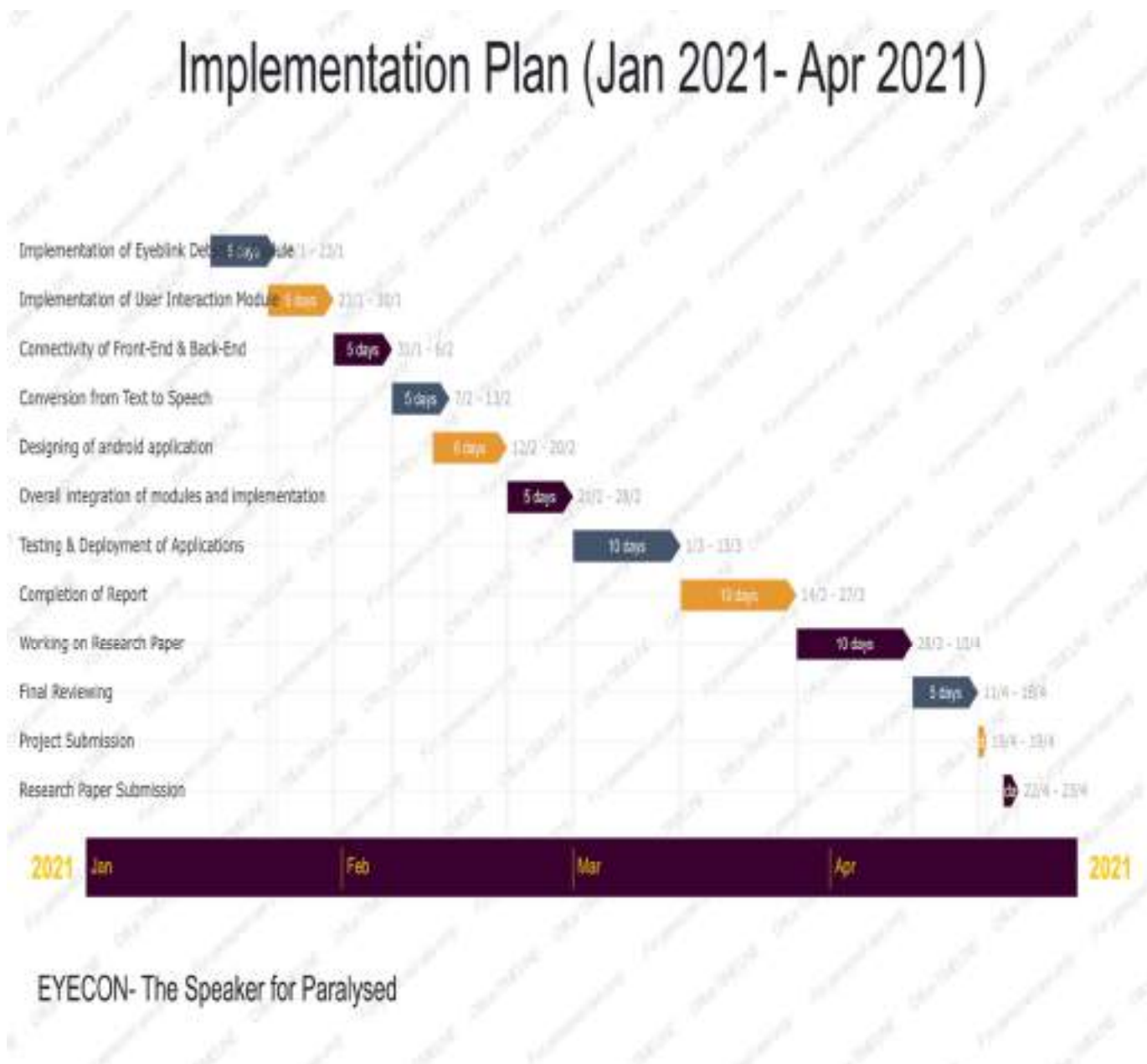


Fig 5.4- Graphical Representation of Project Timeline (Jan 2021- April 2021)

Chapter 6

System Design and Implementation

6.1 System Design

Systems design is the process of defining elements of a system like modules, architecture, components and their interfaces and data for a system based on the specified requirements. It is the process of defining, developing and designing systems which satisfies the specific needs and requirements of a business or organization. A systemic approach is required for a coherent and well-running system. Bottom-Up or Top-Down approach is required to take into account all related variables of the system. A designer uses the modelling languages to express the information and knowledge in a structure of system that is defined by a consistent set of rules and definitions

6.1.1 Flow graph of Blink Detection Module

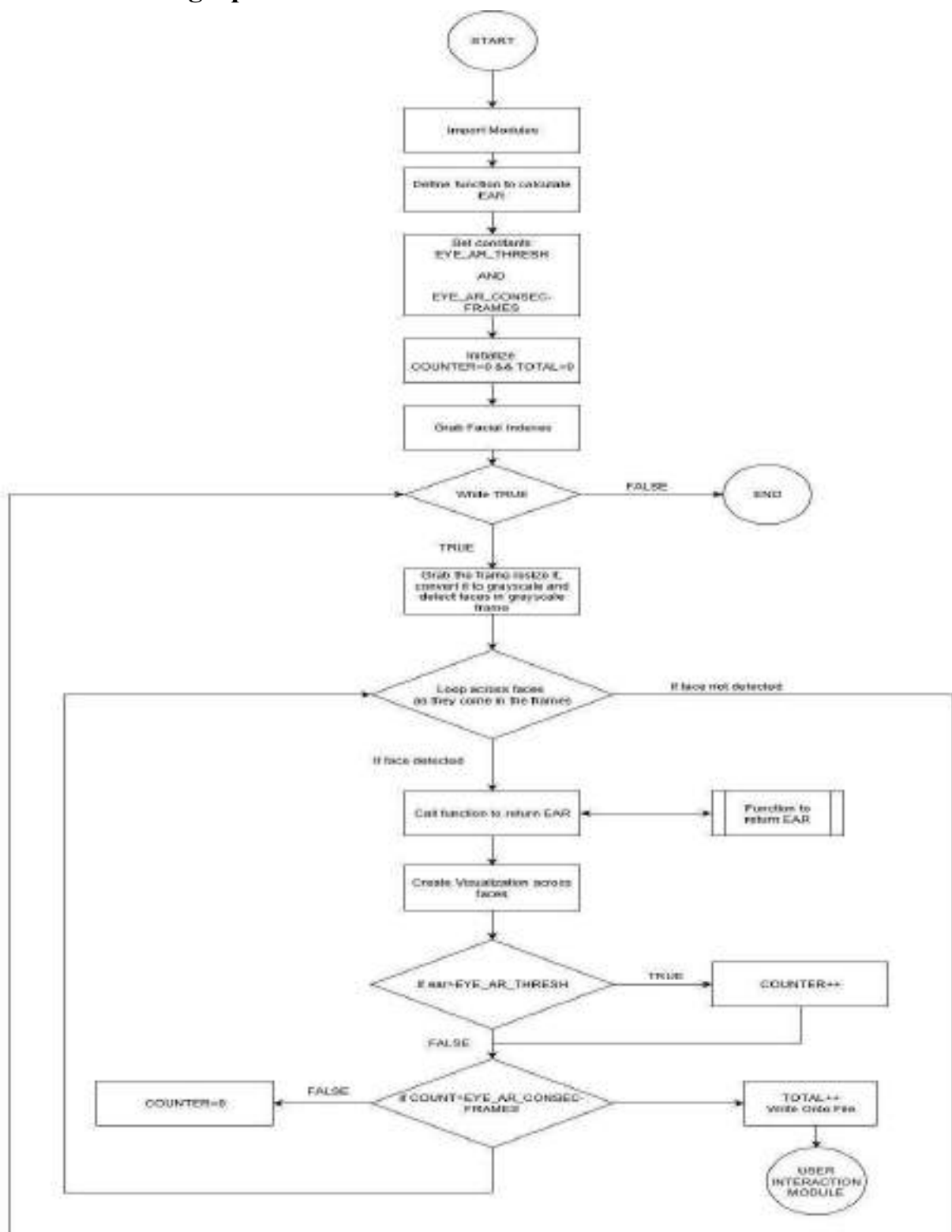


Fig 6.1 – Flow graph of Blink Detection Module

6.1.2 Flow graph of User Interaction Module

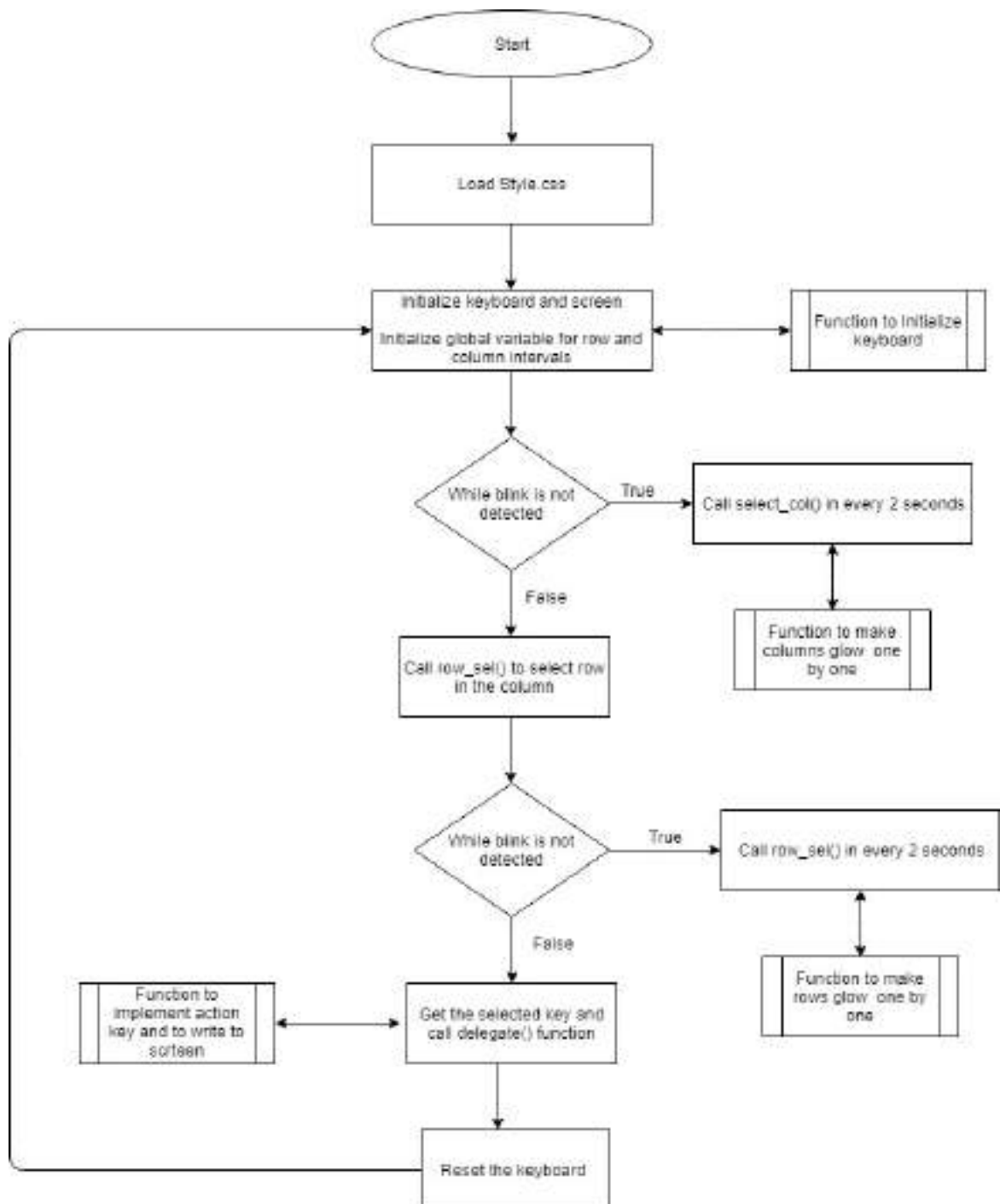


Fig 6.2 – Flow graph of User Interaction Module

6.1.3 Data Flow Diagram

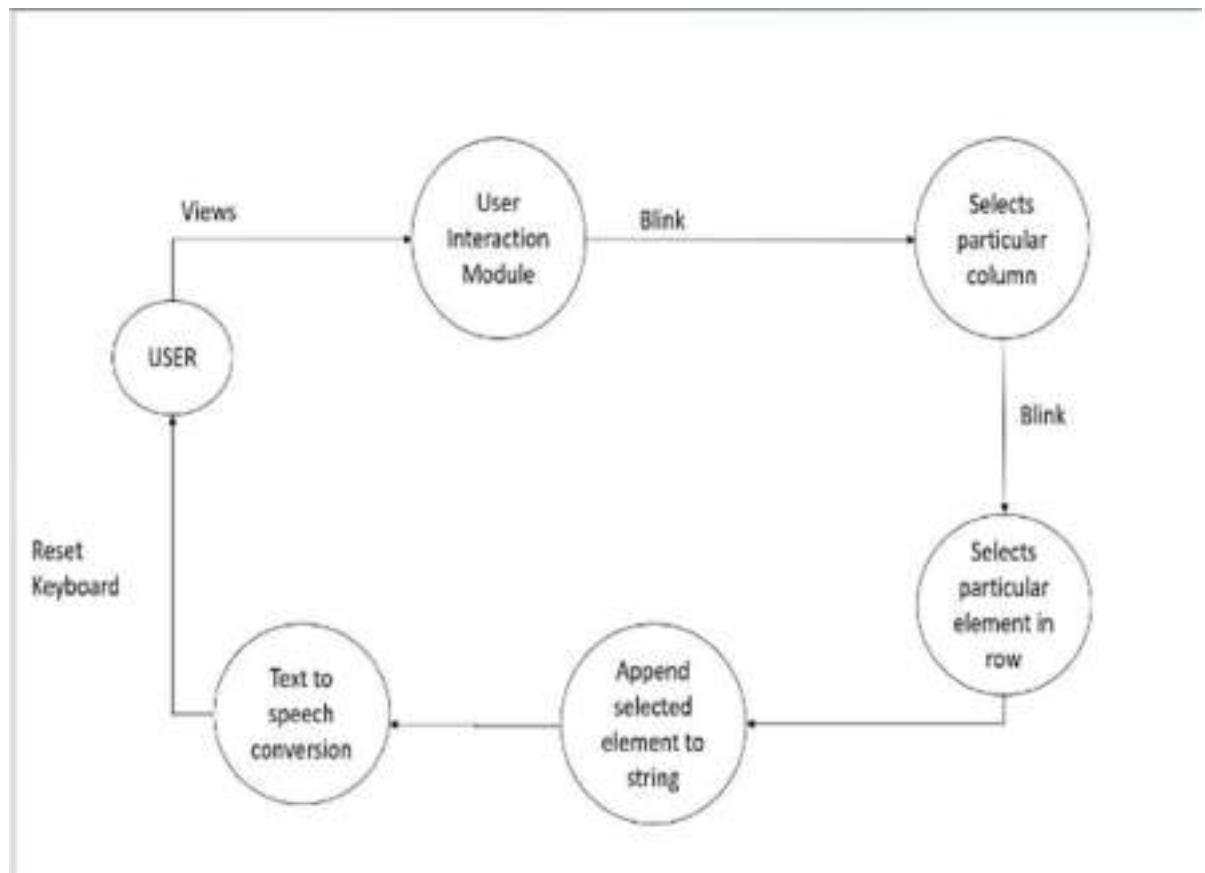


Fig 6.3 – Data Flow diagram

6.1.4 Activity Diagram

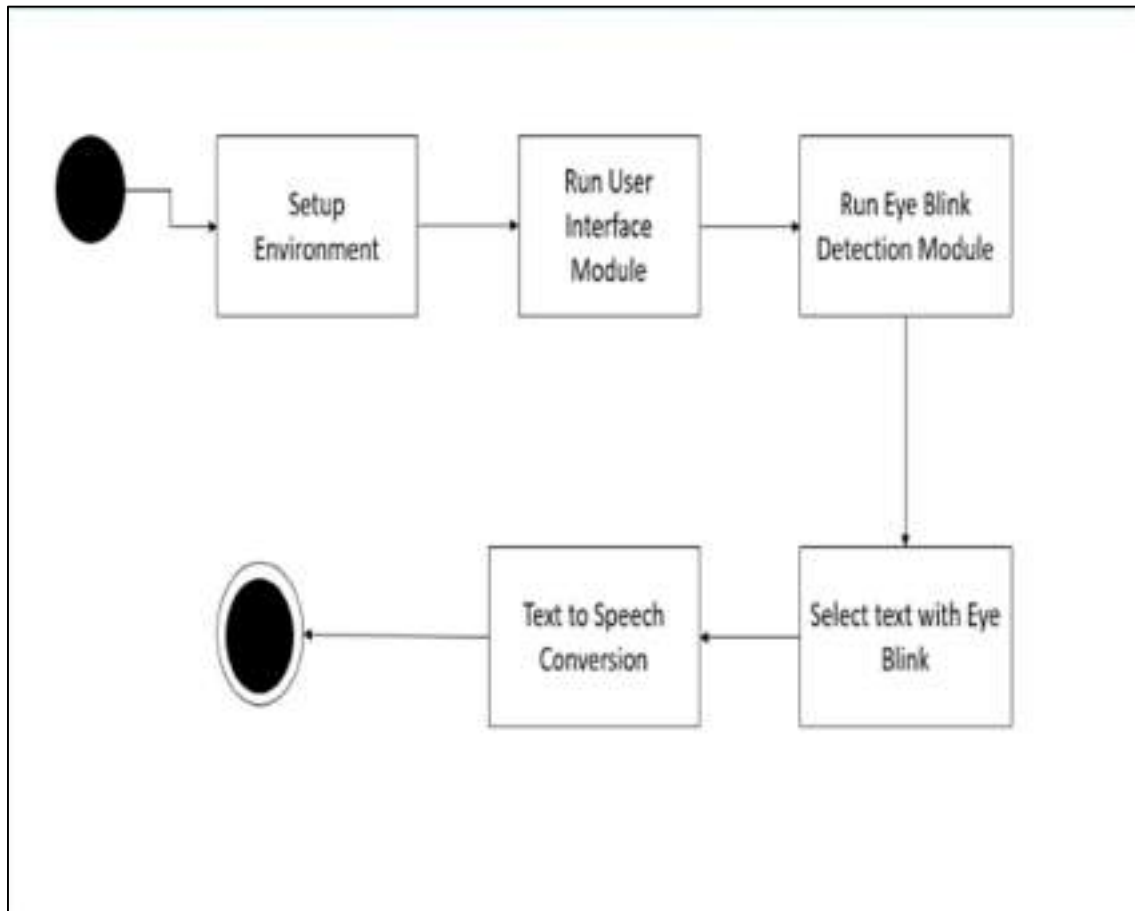


Fig 6.4 – Activity Diagram

6.1.5 Class Diagram

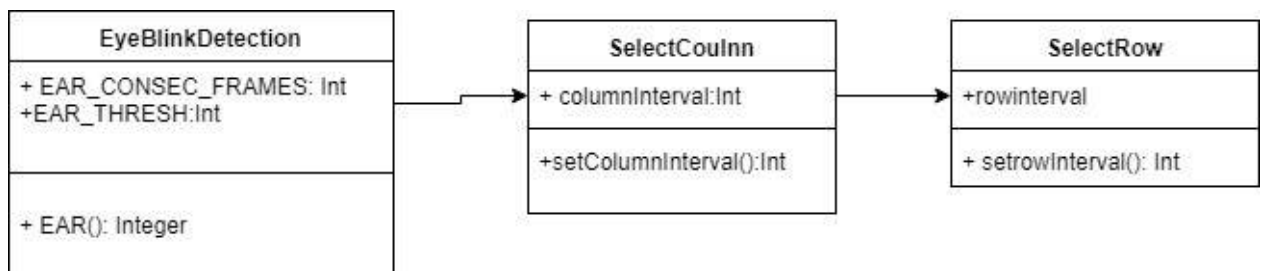


Fig 6.5 – Class Diagram

6.1.6 Use Case Diagram

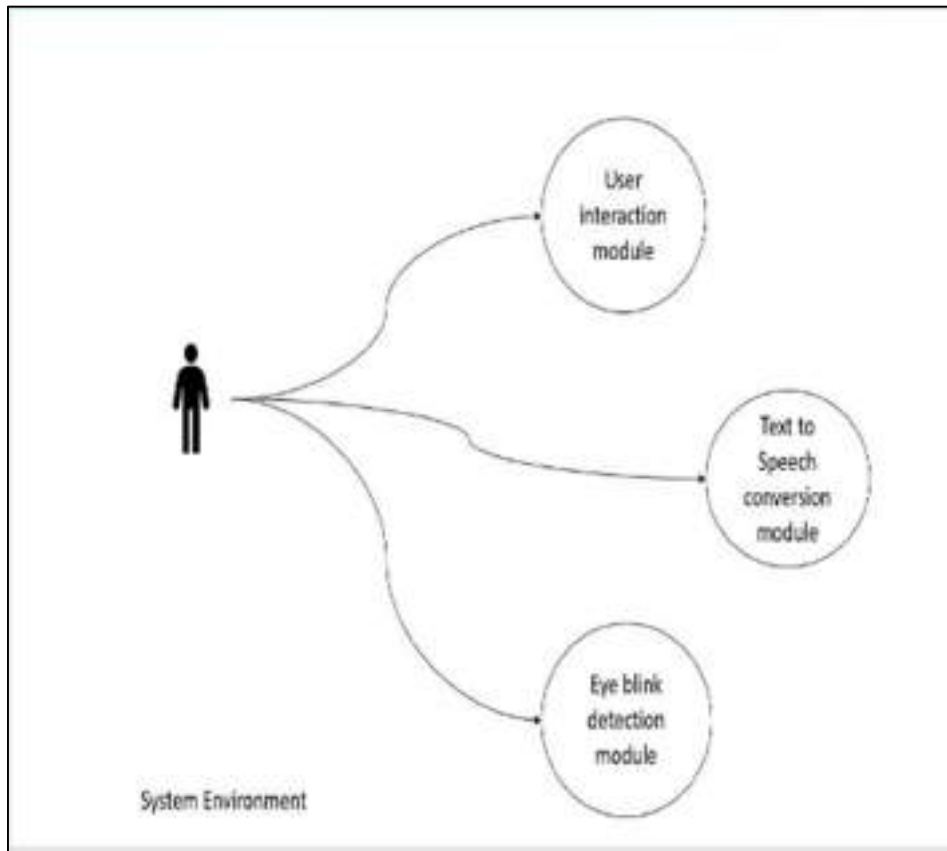


Fig 6.6 – Use Case Diagram

6.1.7 Workflow of Blink Detection Module

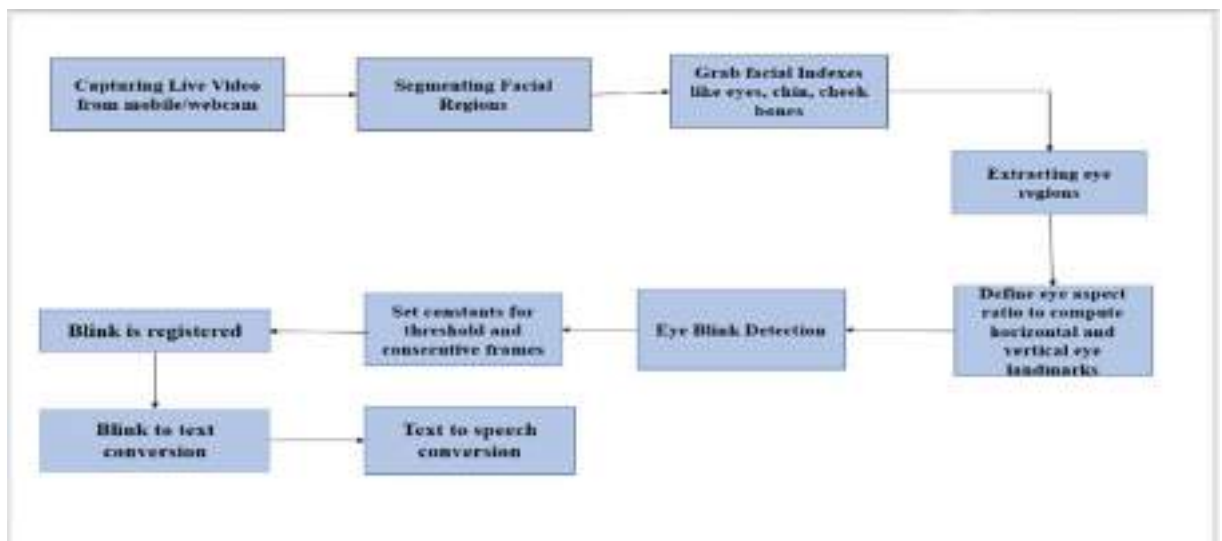


Fig 6.7 – Workflow of Blink Detection Module

6.1.8 Workflow of User Interaction Module

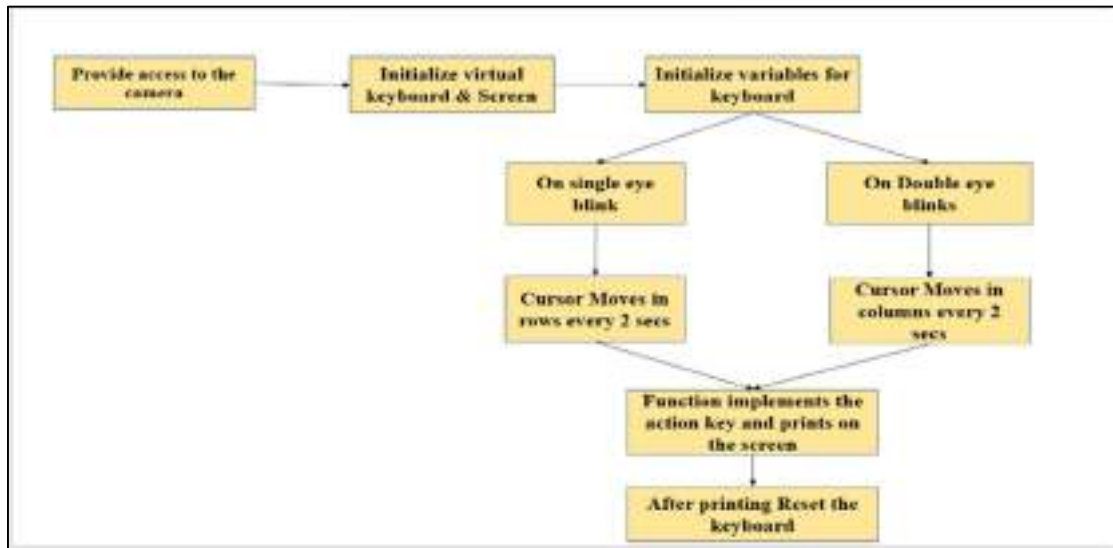


Fig 6.8 – Workflow of User Interaction Module

6.2 Implementation

The project has been split into 3 major modules for the sake of simplicity while implementation. The following subsections will elaborate on all the pre-requisite libraries downloaded, the module implementations and the additional technologies used.

6.2.1 Installation of prerequisite libraries and setting up of Anaconda environment

6.2.1.1 Installation and setting up of Anaconda environment

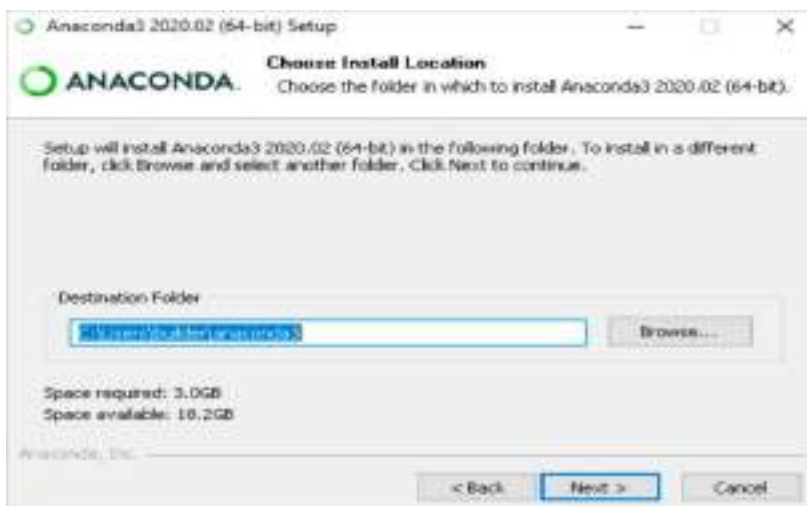


Fig 6.9– Installation of Anaconda

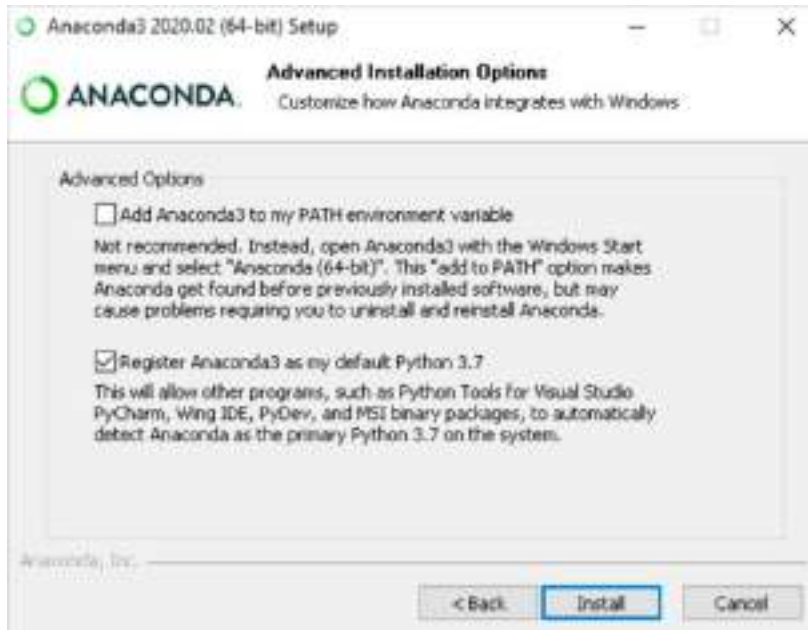


Fig 6.10– Installation of Anaconda



Fig 6.11 – Installation of Anaconda

6.2.1.2 Installation of prerequisite libraries

- A) Numpy – `pip install numpy`
- B) Sklearn - `pip install sklearn`
- C) Flask – `pip install flask`
- D) Pandas – `pip install pandas`
- E) Matplotlib – `pip install matplotlib`
- F) Install Opencv and Dlib - `conda create --name opencv-env python=3.6`

```

C:\Users\mhammad\Anaconda>conda create --name opencv-env python=3.6

Fetching package metadata .....
Building package specifications: ..

package plan for installation in environment C:\Users\mhammad\Anaconda\envs\opencv-env:

The following new packages will be INSTALLED:

  certifi:                2017.11.3-py36h33424ee_0
  pip:                    9.0.1-py36h33424ee_0
  python:                 3.6.5-h33424ee_2
  setuptools:            38.4.0-py36h33424ee_0
  wc:                     1.8-h33424ee_2
  vs2015_runtime:        14.0.25123-h33424ee_2
  wheel:                  0.30.0-py36h33424ee_1
  wincurstern:           0.2-py36h33424ee_0

Proceed [y]/n)?

```

Fig 6.12 – Installation of Opencv

```

C:\Users\mhammad\Anaconda>conda create --name opencv-env python=3.6

Fetching package metadata .....
Building package specifications: ..

package plan for installation in environment C:\Users\mhammad\Anaconda\envs\opencv-env:

The following new packages will be INSTALLED:

  certifi:                2017.11.3-py36h33424ee_0
  pip:                    9.0.1-py36h33424ee_0
  python:                 3.6.5-h33424ee_2
  setuptools:            38.4.0-py36h33424ee_0
  wc:                     1.8-h33424ee_2
  vs2015_runtime:        14.0.25123-h33424ee_2
  wheel:                  0.30.0-py36h33424ee_1
  wincurstern:           0.2-py36h33424ee_0

Proceed [y]/n)?

#
# To activate this environment, use:
# > activate opencv-env
#
# To deactivate an active environment, use:
# > deactivate
#
# * For power-users using bash, you must source
#
C:\Users\mhammad>

```

Fig 6.13 – Installation of OpenCV

```

pip install numpy scipy matplotlib scikit-learn jupyter
pip install opencv-contrib-python
pip install dlib

```

```

import cv2
cv2.__version__
import dlib
dlib.__version__

```

```

C:\Users\adnan\Downloads> pip install opencv-python==3.4.11.0-cp36-cp36m-win_amd64.whl
Processing c:\users\adnan\downloads\opencv-python==3.4.11.0-cp36-cp36m-win_amd64.whl
Installing collected packages: opencv-python
Successfully installed opencv-python-3.4.11.0

C:\Users\adnan\Downloads> pip install dlib
Collecting dlib
Using cached dlib-19.7.0-cp36-cp36m-win_amd64.whl
Installing collected packages: dlib
Successfully installed dlib-19.7.0

C:\Users\adnan\Downloads> python
Python 3.6.4 [AMD64, Intel(R) Core(TM) i7-6700 CPU @ 3.40GHz] on win32
Type "help", "copyright", "credits" or "license()" for more
>>> import cv2
>>> cv2.__version__
'3.4.11'
>>> import dlib
>>> dlib.__version__
'19.7.0'
>>> exit()

```

Fig 6.14 – Installation of Dlib

6.2.2 Implementation of Eye Blink Detection Module

A) Information about the module-

We have made use of Dlib, which is a modern C++ toolkit containing machine learning algorithms and equipment for creating complicated software. This Dlib library was then constructed on top of python for our software. Face detection was accomplished with the use of dlib. After that, the eye area was localized from the facial area with the use of Eye Aspect Ratio (EAR). Each eye was represented with the aid of a set of 6(x,y) coordinates, starting at the left corner of the eye and then working around in a clockwise direction. These form 2 sets of vertical landmarks and a set of horizontal landmarks. These landmarks are denoted in the following picture-

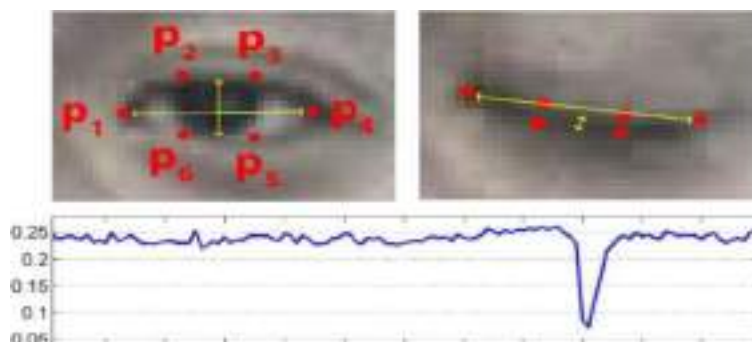


Fig 6.15 – Calculation of EAR co-ordinates

The Equation that helps derive the relationship between these coordinates can be given as-

$$\text{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|},$$

Fig 6.16 – Calculation of EAR co-ordinates

Where, $p_1 \dots p_6$ are 2D facial landmark locations. In this equation, the numerator computes the distance between vertical landmarks while the denominator computes the distance between horizontal landmarks. The eye aspect ratio is about steady whilst the eye is open however will swiftly fall to 0 while a blink is taking place. As indicated in figure 3, while the eye is completely open, the Eye- Aspect Ratio is huge and relatively consistent over time. Once someone blinks, it decreases dramatically drawing close 0. The bottom figure plots a graph of the eye aspect ratio over time for a video clip. As we can see, the eye element ratio is consistent, then swiftly drops near 0, then will increase again, indicating a single blink has taken place.

Now in order to determine if a blink has taken place, we've got defined variables- threshold consecutive frames. If the eye aspect ratio falls under a certain threshold and then rises above the threshold, then we register a blink—the EYE AR THRESH is this threshold value. Another constant is EYE AR CONSEC FRAME which is set to a selected value which was three for our application indicates that 3 successive frames with an eye aspect ratio much less than EYE AR THRESH must happen in order for a blink to be registered. Another variable specifically COUNTS and TOTAL had been initialised. The TOTAL variable gave the variety of blinks detected whilst this system was running. The COUNT was used to test whether or not the EYE AR CONSEC FRAME constraint was met or not.

B) Screenshots of Implementation

```
def eye_aspect_ratio(eye):
    # compute the euclidean distances between the two sets of
    # vertical eye landmarks (x, y)-coordinates
    A = dist.euclidean(eye[1], eye[5])
    B = dist.euclidean(eye[2], eye[4])

    # compute the euclidean distance between the horizontal
    # eye landmark (x, y)-coordinates
    C = dist.euclidean(eye[0], eye[3])

    # compute the eye aspect ratio
    ear = (A + B) / (2.0 * C)

    # return the eye aspect ratio
    return ear
```

Fig 6.17 – Function EAR

```
# initialize dlib's face detector (HOG-based) and then create
# the facial landmark predictor
print("[INFO] loading facial landmark predictor...")
detector = dlib.get_frontal_face_detector()
predictor = dlib.shape_predictor(r"C:\Users\deepi\Downloads\eyeCon3\shape_predictor_68_face_landmarks.dat")

# grab the indexes of the facial landmarks for the left and
# right eye, respectively
(lStart, lEnd) = face_utils.FACIAL_LANDMARKS_IDXS["left_eye"]
(rStart, rEnd) = face_utils.FACIAL_LANDMARKS_IDXS["right_eye"]
```

Fig 6.18 – Loading Face Detector

```
face_cascade = cv2.CascadeClassifier(r"C:\Users\deepi\Downloads\eyeCon3\static\haarcascade_frontalface_default.xml")
eye_cascade = cv2.CascadeClassifier(r"C:\Users\deepi\Downloads\eyeCon3\static\haarcascade_eye_tree_eyeglasses.xml")
cap = cv2.VideoCapture(0)
count = 0
while 1:
    ret, img = cap.read()
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(gray, 1.3, 5)
    for (x,y,w,h) in faces:
        cv2.rectangle(img,(x,y),(x+w,y+h),(255,255,255))
        roi_gray = gray[y:y+h, x:x+w]
        roi_color = img[y:y+h, x:x+w]
        roi_grayLeft = gray[y:y+h, x:x*w/2]
        roi_grayRight = gray[y:y+h, x*w/2:x*w]
        leftEye = eye_cascade.detectMultiScale(roi_grayLeft)
        rightEye = eye_cascade.detectMultiScale(roi_grayRight)
        if len(leftEye) > 0 and len(rightEye) > 0:
            count = count + 1
            print(count)
            file = open(r"C:\Users\deepi\Downloads\eyeCon3\eye.txt", "w")
            file.write("1")
            file.close()
```

Fig 6.19 – Capturing Eye Blinks

6.2.3 Implementation of User Interaction Module

A) Info about the module

User Interaction Module The implementation of the user interaction module consisted of the subsequent steps- 1) Initializing the keyboard, 2) Setting interval for the column, 3) Setting interval for row upon column selection 4) Blink detection. We have made use of JavaScript code for the initialization of the keyboard. It consisted of the `init()` function which specified if the CAPS should be on or not. After the initialization of the keyboard, we set the column interval that highlighted a column in every 2- second delay. If there has been a blink interrupt at some point of this time then that column was selected after which we started the row interval for that specific column. For the selection of the row, `selectRow()` function was implemented which after detection of the blink selected a particular key and displayed it. In order to fetch detection records from the Blink detection module, we have used AJAX to send requests to the Flask that is linked to the Blink detection module using FIFO. In each 1ms AJAX dispatched a request to look if a blink was detected or not. Once the blink was detected we had to change the front end. We made use of the `detection ()` function that did the essential changes. This function would either write to the display screen the key selected if it was a number or an alphabet or would perform an operation if an action key had been selected. An action key included the Backspace, Caps, and Tab. The detection function was used to clear the intervals and reset the keyboard to its preliminary stage.

B) Screenshots of Implementation

```

129
130  //initializing key board
131  var keys=['BACKSPACE','a','b','c','d','e','f','g','h','i','j',
132  'SPACE','k','l','m','n','o','p','q','r','s','t','CAPS','u',
133  'v','w','x','y','z','RESET'];
134
135

```

Fig 6.20 – Function for Virtual Keyboard


```

70 //for functional keys
71
72 function delegate(key)
73 {
74     if(key=="BACKSPACE")
75     {
76         text=text.substr(0,text.length-1);
77         document.getElementById("final").textContent=text;
78     }
79     else if(key=="SPACE")
80     {
81         text+=" ";
82         document.getElementById("final").textContent=text;
83     }
84     else if(key=="CAPS")
85     {
86         $(".li.buttons").remove();
87         init(traw);
88     }
89     else if(key=="RESET")
90     {
91         reset2();
92     }
93 }
94

```

Fig 6.21– Function for special keywords

```

212 //select row
213 function selectflow(col_nue)
214 {
215     $(attr).css("background","black");
216
217     var array1=[](col_nue);
218     var i=0;
219     var mod=array1.length;
220     var cnt=0;
221
222     row_tim=setInterval(function(){
223
224         j=i%mod;
225
226         if(j==0)
227             array1[j-1].style.backgroundColor="black";
228         else if(j==mod-1)
229             array1[j+mod-1].style.backgroundColor="black";
230
231         array1[j++].style.backgroundColor="grey";
232
233         i++;
234         if(j==0)
235         {
236
237             ans=array1[j-1].innerHTML;
238
239             document.getElementById("input").textContent=ans;
240             ans="";
241             rowFlag=true;
242         }
243     },2000);
244 }
245

```

Fig 6.22 – Function to calculate row intervals

6.2.4 Implementation of Text to Speech Conversion Module

A) Information about the module-

In order to add a text to speech feature on our web page using JavaScript, we have used a Web Speech API, to synthesize text and convert it into speech. `SpeechSynthesisUtterance` is the interface in which we actually create the speech or utterance using the text provided, setting a language type, volume, pitch of the voice, rate of speech etc. The function's name is `TexttoAudio`. This JavaScript method consists of different properties like language, pitch, rate, volume, text and voice. Once the text is passed in this function, it is read and further passed on to the speech synthesis interface. Then as per how the values have been set, the properties of this method, speech output would be produced. The gender of the speech would depend on the particular system's default voice.

B) Screenshots of Implementation

```
247     function textToAudio() {  
248         let msg = text;  
249  
250         let speech = new SpeechSynthesisUtterance();  
251         speech.lang = "en-US";  
252  
253         speech.text = msg;  
254         speech.volume = 1;  
255         speech.rate = 1;  
256         speech.pitch = 1;  
257  
258         window.speechSynthesis.speak(speech);  
259     }
```

Fig 6.23– Function for Text to Speech Conversion

6.2.4 Implementation of Website



Fig 6.24– Website window

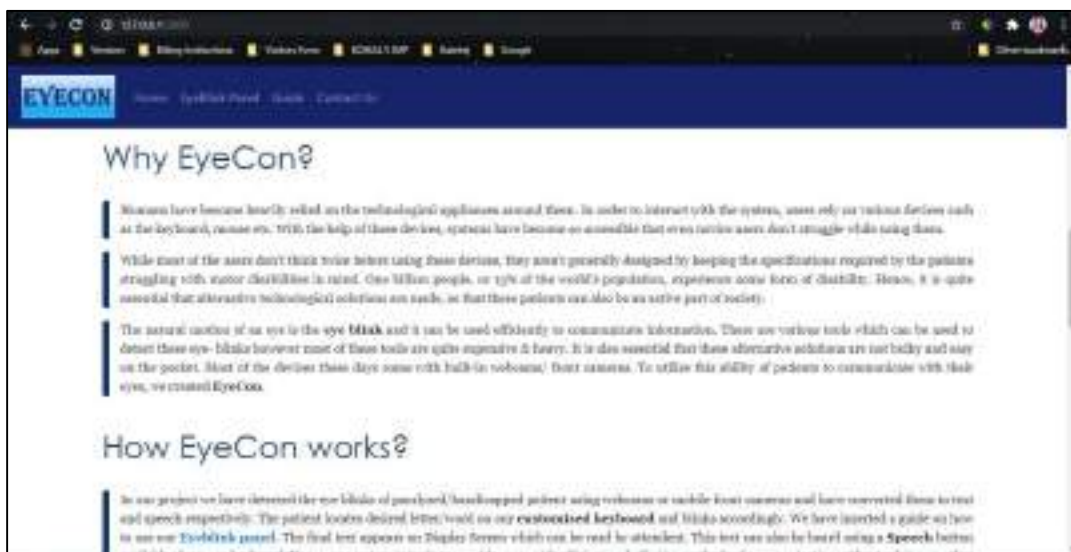


Fig 6.25 – Website window



Fig 6.26 – Launch the panel



Fig 6.27 – User Instructions



Fig 6.28– Contact page

6.2.6 Linking of Website with User Interface

```
@app.route("/eyeblickpanel")
def eyeblickpanel():
    return render_template("eyeblickpanel.html")

@app.route("/index1")
def index1():
    return render_template("index1.html")

@app.route("/guide")
def guide():
    return render_template("guide.html")

@app.route("/contact")
def contact():
    return render_template("contact.html")

@app.route('/process', methods=['POST'])
```

Fig 6.29 – Linking the website with Flask



Fig 6.30 – User Interface

Chapter 7

Results and Discussion

7.1 Results

7.1.1 Results of the Eye Blink Detection Module

In this module, the facial region was isolated by making use of Haar Cascade Classifier. Landmark detection was performed upon which the eye region was located. The eye region was further identified by calculating the eye aspect ratio and the eye blinks were noted. The input given to this module was the face of the user which was taken as the input via the webcam. The result obtained was the eye blinks occurring which were successfully noted and the number of eye blinks taking place were noted in python file

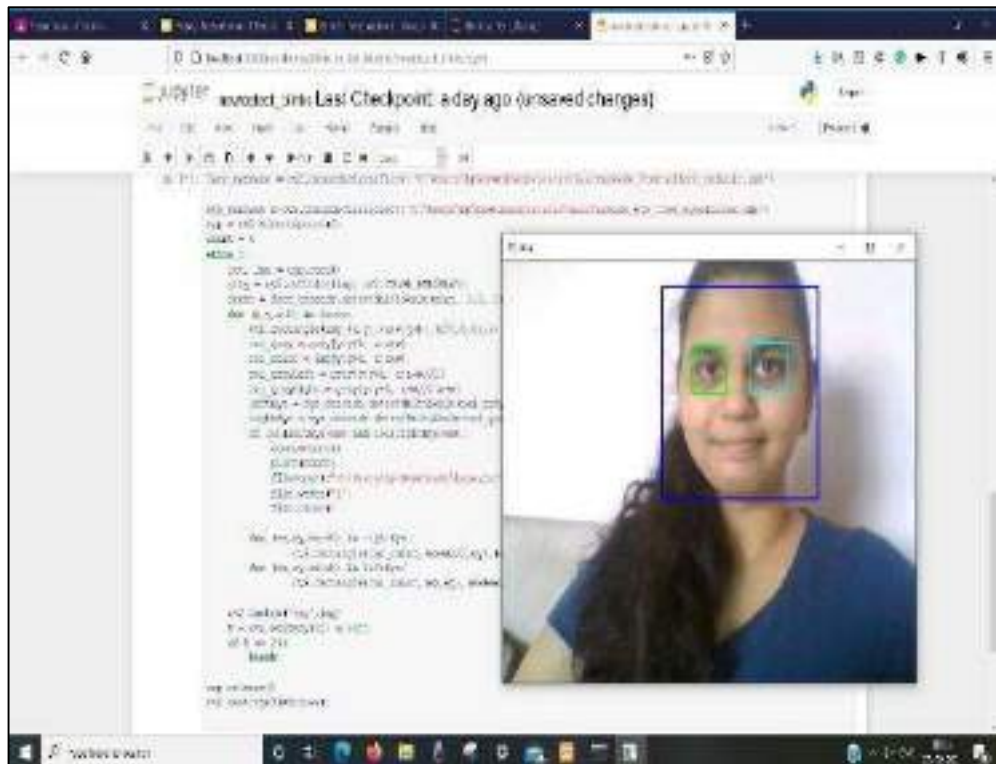


Fig 7.1 –Results of Blink Detection Module

7.1.2 Results of the User Interaction Module

In this module, the eye blinks that have been identified in the previous module are used to navigate across the keyboard which is present on the screen. The navigation takes place across the keyboard with the navigator moving column-wise first and then row-wise after a particular column has been selected. The input given to this module was the eye blinks that were being captured via the live stream video. The result obtained was the text which had been typed with the help of the blinks.



Fig 7.2 –Results of User InteractionModule



Fig 7.3 –Results of User Interaction Module

7.1.2 Results of the Text to Speech Conversion Module

In this module, the main aim was to convert the text that the user has formulated with the

help of his eye blinks by navigating across the keyboard to speech. The input given to this module was the text that the user had typed with the help of blinks. The result obtained was the text that was converted to speech.



Fig7.4–Results of Text to Speech conversion Module



Fig 7.5 –Results of Text to Speech conversion Module

7.2 Test Cases

7.2.1 Testing of Eye Blink Detection Module-

Eye blink detection testing has been done to check the correctness of the eye blink module which was implemented. The main goal of eye blink detection module was to successfully isolate the eye part from the facial region and capture the eye blinks of the users. The following table shows the various test cases for eye blink detection module. Eye blink detection is used to eventually select alphabets for the keyboard. The number of milliseconds that the number of blinks was done and should generate a blink signal is 1 second and was set as the threshold. We have also checked if the system is able to distinguish between different kinds and colours of eyes. The following table shows the various test cases for eye blink detection module-

Sr. No	Test Case Description	Input	Expected Output	Actual Output	Result
1.	Eyes Shut	10 pairs of eyes at different instances.	No blinks should be detected	Blinks were not detected	PASS
2.	Eye Detection	Brown Eyes	Blinks should be detected	Blinks were detected	PASS
3.	Eye Detection	Black Eyes	Blinks should be detected	Blinks were detected	PASS
4.	Eye Detection	Grey Eyes	Blinks should be detected	Blinks were detected	PASS
5.	Eye Detection	Eyes containing cataract	Blinks should be detected	Blinks were not detected	FAIL

6.	Eye Detection	Cataract operated eyes	Blinks should be detected	Blinks were detected	PASS
----	---------------	------------------------	---------------------------	----------------------	------

Table 7.1- Tabular Representation of Eye Blink Detection Test cases

7.2.2 Testing of User Interaction Module

User Interaction Module testing had been implemented to check the correctness of the functioning of the module which was implemented. The main aim of this module was to read the eye blinks as input and to use them for navigating across the keyboards. Parameters were defined for this navigation. Initially, a blinker started its motion on the keyboard from left to right in a cyclic pattern alongside the columns present on the keyboard. Upon, blinking of the eyes once, a particular column on the keyboard was selected and then the blinker began its navigation in the rows of that particular column. The accuracy was then determined by the number of words correctly typed. The following table illustrates the functioning of this module-

Sr. No	Test Case Description	Input	Expected Output	Actual Output	Result
1.	Initialization of the keyboard.	The clicking of the launch button on the website.	The keyboard should have been initialized.	The keyboard was initialized.	PASS
2.	Navigation of the blinker across columns.	Initialization of keyboard.	The blinker should have navigated across columns.	The blinker navigated across columns.	PASS
3.	Navigation of the blinker across rows.	Selecting a particular column.	Navigation of rows upon	Blinker navigated across the rows on column selection.	PASS

			column selection.		
4.	Typing of alphabets using eye blinks	The eye blinks of the user.	Selection of alphabets on the keyboard	The alphabets were correctly selected with the help of blinks	PASS
5.	Typing of numbers using eye blinks	The eye blinks of the user	Selection of numbers on the keyboard	The numbers were correctly selected with the help of blinks	PASS
6.	Typing of special keys using eye blinks	The eye blinks of the user	Selection of special keys on the keyboard	The special keys were correctly selected with the help of blinks	PASS

Table 7.2- Tabular Representation of User Interaction Test cases

7.2.3 Testing of Text to Speech Conversion Module-

Text to Speech Conversion Module testing had been implemented to check the correctness of the functioning of the module which was implemented. The main objective of this module was to correctly translate to speech, the text that was given as input. The following table illustrates the different test cases tried for the same-

Sr. No	Test Case Description	Input	Expected Output	Actual Output	Result
1.	Functioning of the speak button	Clicking on the speak button.	Translation of the text to speech	The text was converted to speech on clicking of speak button	PASS
2.	Conversion of the alphabet to speech	Clicking on the speak button.	Alphabet to be converted to Speech.	The speech was obtained from the input alphabet.	PASS
3.	Conversion of numbers to speech	Clicking on the speak button.	Number to be converted to Speech	The speech was obtained from the input number.	PASS
4.	Conversion of words to speech	Clicking on the speak button.	Words to be converted to Speech	The speech was obtained from the input words.	PASS
5.	Conversion of random string (consisting of alphabets and numbers) as a coherent word in speech format.	Clicking on the speak button.	Speech to be obtained as coherent words.	The the speech wasn't read coherently.	FAIL
6.	Conversion of random string (consisting of alphabets and numbers) as a randomly read string in speech format.	Clicking on the speak button.	String to be converted to Speech in non-coherent form.	The speech was obtained of in the form of a random string.	PASS
7.	Conversion of sentences to speech	Clicking on the speak button.	Sentences to be converted to Speech	The speech was obtained of the input sentences	PASS

Table 7.3- Tabular Representation of Text to Speech Conversion Test cases

7.2.4 Functional Testing-

In this testing, all the above-mentioned modules which have now been integrated together were tested as one whole system. So, the main job of this whole system was to take eye blinks as input from the user which were then used to navigate on the keyboard as a part of the user interaction module, with which words were formed which were shown on the output screen and this was eventually converted to speech. The following table shows the working of the same-

Sr. No	Test Case Description	Input	Expected Output	Actual Output	Result
1.	Intended Text	Aditi Godbole	Aditi Godbole	Aditi Godbole	PASS
2.	Intended Text	Deepika Rao	Deepika Rao	Deepika Rao	PASS
3.	Intended Text	Komal Pawoolkar	Komal Pawoolkar	Komal Pawoolkar	PASS
4.	Intended Text	Comp Eng	Comp Eng	Comp Eng	PASS
5.	Intended Text	NU768	NU768	NU768	PASS
6.	Intended Text	qazwsx123	qazwsx123	qazwsx123	PASS

Table 7.4- Tabular Representation of Functional Testing Test cases

7.3 Discussion

Live stream videos were taken as inputs during the testing of this module. The high-speed webcam was used to read the video as input from the user. A total of 10 people participated during the trials of the system. The output was reviewed by three individuals. During the review, the individuals watched the participants from aside and noted each time during which

the participants blinked and whether the system registered it as a blink or a miss. Every time, the participant blinked and the system correctly identified it, it was classified as a blink and for the occasions on which the system failed to identify the blink, it was noted as a miss. If the system detected a blink and if the participant had not performed the activity of blinking, it was classified as a False Positive.

Now, humans perform blinks almost involuntarily in many instances. Hence, in order to distinguish the spurious blink from an actual one, we have defined a threshold value of 3 seconds. Another variable of consecutive frames was also defined. If a blink did take place for this duration, for a consecutive number of frames, only then it was actually classified as an actual blink and the rest were ignored as spurious blinks. **A total of 1000 blinks were noted for testing purpose. Only a total of 5% blinks out of 1000 blinks i.e., 50 blinks were missed and a mere 1% of blinks i.e. 10 blinks were falsely detected. Overall, the eye blink detection module performed with an accuracy of 94%.**

Chapter 8

Conclusion

As a part of this project, we had proposed a system which would take eye blinks of the patients as an input and would effectively transform them to text and speech. In order to perform the same, we had divided this project into 3 modules namely Eye Blink detection Module, User Interaction Module and Text to Speech Module. As a part of the Eye Blink Detection module, we have captured the eye blinks of the patient and the same was done by the use of Haar Cascade Classifiers and calculation of Eye Aspect Ratio. Based on the threshold limits and consecutive frames a blink was registered. In the User Interaction Module, we have created a keyboard and an output screen. The navigation across the keyboard was performed by a blinker column wise on first blink and row wise on second blink. In Text to Speech module a function TexttoAudio was defined in JavaScript to convert the obtained text from user interaction module into speech. Various parameters like rate, pitch, volume etc. could be adjusted as a part of this module.

The results for each of the module are summarized below-

- **Eye Blink Module** - Eye blinks occurring were successfully recorded and noted. A total 1000 blinks were noted for testing purpose. 5% blinks out of 1000 i.e., 50 blinks were missed and 1% i.e 10 blinks were falsely detected. Accuracy achieved in this module is 94%.
- **User Interaction Module** - The text typed with the help of the blinks is the output. The accuracy was analysed on the basis of navigation of blinker across rows and columns and the words being correctly rendered.
- **Text to Speech Conversion Module** - The text was converted to speech in this module. Accuracy was gauged on the basis of words correctly translated to speech.
- **Website** - Our website is hosted using Flask and Anaconda. HTML and CSS are used to design the frontend. The Eyeblink interface designed using JavaScript has been integrated and is working successfully.

We have performed test cases for all the three modules for various scenarios out of which one case failed in eye blink detection module and one case failed in text to speech module. This led us to discover the shortcomings of the project and hence come up with potential areas which can be improved upon in future. Thus we have successfully curated a system which is cost efficient and is a very effective tool for the user.

Future Scope

As we know no system is perfect and there are always methods that we can undertake to enhance our system which brings us to discuss the shortcomings of the project and the possible ways in which it can be enhanced which we will explain under the future scope.

1. Our project requires that the face is fully visible, only then the camera would take it as an input. The system would render inefficient if the face is partially or obliquely visible. Hence, a future scope could be that the system continues effortlessly even if the face is partially visible.
2. The model also requires that the face is sufficiently illuminated & it cannot function when there is dim or gloomy lighting. Efforts can be made to mitigate this situation in future endeavours.
3. The model should also be modified so that its functioning can also be extended to patients having cataract so that it ensures better inclusivity.
4. Presently, our keyboard only has basic alphabets and numbers. In future, additional commonly used words like Thank you, Please, Water etc. can be added to ensure smooth functioning. The auto-suggest feature can also be included for a better user experience.
5. Currently, the text & speech support is provided only in the English language. It can be later extended to provide support in various regional & international languages.

6. Furthermore, the text to speech history could be saved in a database and displayed under a history tab on the website so that the patients could take a look at the previous conversations if required.

References

- 1) Su, M., Yeh, C., Lin, S., Wang, P., & Hou, S. (2008, July). An implementation of an eye-blink-based communication aid for people with severe disabilities. In *2008 International Conference on Audio, Language and Image Processing* (pp. 351-356). IEEE.
- 2) Soukupova, T., & Cech, J. (2016, February). Eye blink detection using facial landmarks. In *21st computer vision winter workshop, Rimske Toplice, Slovenia*.
- 3) Mukherjee, K., & Chatterjee, D. (2015, January). Augmentative and alternative communication device based on eye-blink detection and conversion to morse-code to aid paralyzed individuals. In *2015 International Conference on Communication, Information & Computing Technology (ICCICT)* (pp. 1-5). IEEE.
- 4) Goyal, K., Agarwal, K., & Kumar, R. (2017, April). Face detection and tracking: Using OpenCV. In *2017 International conference of Electronics, Communication and Aerospace Technology (ICECA)* (Vol. 1, pp. 474-478). IEEE.
- 5) Sharifi, M., Fathy, M., & Mahmoudi, M. T. (2002, April). A classified and comparative study of edge detection algorithms. In *Proceedings. International conference on information technology: coding and computing* (pp. 117-120). IEEE.
- 6) Abu-Faraj, Z. O., Sleiman, H. C. B., Al Katergi, W. M., Heneine, J. L. D., & Mashaalany, M. J. (2008). A rehabilitative eye-tracking based brain-computer interface for the completely locked-in patient. In *Encyclopedia of Healthcare Information Systems* (pp. 1153-1160). IGI Global.

- 7) KARKI, M. A., TIMILSINA, M. A., REDDY, M. V. I. K., HEGDE, M. S., & GUIDE,
- 8) M. (2018). CONVERTING INTENT TO TEXT USING REAL-TIME EYEBLINK DETECTION.
- 9) Mohammed, A., & Anwer, S. (2014). Efficient Eye Blink Detection Method for the disabled-helping domain. Inter. Jour. of Adv. In: Computer Science and Applications.
- 10) A REVIEW ON SMART EYE BLINK SOLUTION FOR MND PATIENT USING PYTHON", International Journal of Science & Engineering Development Research (www.ijedr.org), ISSN:2455-2631, Vol.4, Issue 3, page no.361 - 364, March-2019, Tambe Samreen Mohammed , Rajeshwari P.
- 11) Namratha, K. S., Jeevitha, B. S., Sneha, P. H., & Veda, D. Eye blink to voice format communication for paralyzed patients.



PLAGIARISM SCAN REPORT

Date April 27, 2021

Exclude URL: NO



Unique Content **93%**

Plagiarized Content **7%**

Paraphrased Plagiarism **0**

Word Count 395

Records Found 1

CONTENT CHECKED FOR PLAGIARISM:

Users usually rely on various devices such as a keyboard, mouse etc. to communicate with a system. These devices make the system easily accessible and hence even novice users don't find these systems daunting. However, people suffering from motor disabilities or impairment cannot use these devices as they are not customized keeping their needs in mind. Hence, an alternative solution is needed to facilitate glitch less functioning for the impaired. Various researches are being conducted across the globe to overcome this situation. While undertaking the development of such devices, it is critical to consider various aspects such as avoiding the use of heavy and bulky instruments, minimizing the use of manual control, avoiding specialized instruments, featuring real-time performance etc. Previous methods for eye blink detection estimate eye state as either open or closed or track eye closure events. Paralyzed people cannot control muscle function in one or more muscle groups.

This condition is usually caused by diseases like ALS, multiple sclerosis etc. or even stroke. The Locked-In Syndrome ie. LIS is a form of paralysis in which the patients lose control of nearly all voluntary muscles. Apart from eye blinks and eye movement, they are unable to control any other part of their body which renders them inefficient to communicate. However, the patients having LIS are still cognitively aware i.e. thoughts and ideas are forming in their brain. These patients heavily rely on eye blinks for communication purposes. Usually, the nurses or caretakers are trained and well versed in interpreting and decoding their blinks. However, they cannot be understood by laymen and hence in the absence of nurses or caretakers, LIS patients have no means of self- expression.

In order to facilitate a means of expression for LIS patients, we propose a robust system which will facilitate the patients to convert their expression to text and speech. This project will work by taking the eye blinks of the patients as input through webcam and front cameras of mobile phones and would effectively convert them to the text which would be displayed at the bottom of the screen. This text will be converted to audio which will provide ease of accessibility and ensure that the patient is heard out at all times. This project primarily focuses to provide a means of communication to people of all classes without having to worry about financial aspects or cumbersome hardware

MATCHED SOURCES:

[github.com](#) - 4% Similar [Compare](#)

<https://github.com/geekayush/blink>

Report Generated on **April 27, 2021** by prepostseo.com



PLAGIARISM SCAN REPORT

Date April 27, 2021

Exclude URL: NO



Unique Content **100%**

Plagiarized Content **0%**

Paraphrased Plagiarism **0**

Word Count 272

Records Found 0

CONTENT CHECKED FOR PLAGIARISM:

Oculography methodology is employed to create communication between patient and caretaker. The image process module incorporates digital camera and therefore the eye movement- image is captured with OpenCv to get the coordinate of eye ball. The system permits the patients to speak with caretaker victimisation blink pattern converted into voice format. The sensible resolution for image process is obtained by Python programming with OpenCV.

: Existing Technologies

: MindScribe

MindScribe is a system which interprets brain waves specifically designed for people suffering from Locked-In Syndrome (LIS). This device provides three different modes for communication. The first one is expression mode to identify and share emotional state. The second is a yes/no response mode and third is the typing mode. A company named Neurosky has developed this. MindScribe simply has a single headset with sensors that touch the forehead, measuring brain activity.

2.2.2 Eye Control

The EyeControl is the first wearable, screenless, assistive communication device. AI- powered eye-tracking technology enables remote communication between ventilated ICU patients and medical staff via the nurse station. A head-mounted infrared camera tracks the eye movements and sends the information to a small processing unit which translates the movements into communication. The bone conduction component provides audio feedback to the user before the communication is

transmitted to the output speaker or connected Bluetooth device, all without the need for a screen.

Shortcomings Of Existing Technologies

Many of the existing technologies employ the use of specialized devices.

These devices are generally not portable.

Sometimes these devices can cause discomfort.

Many of the topics researched have used features like Haar cascades to detect faces, which do not offer real time eyeblink detection.

MATCHED SOURCES:

Report Generated on **April 27, 2021** by prepostseo.com



PLAGIARISM SCAN REPORT

Date April 27, 2021

Exclude URL: NO



Unique Content **100%**

Plagiarized Content **0%**

Paraphrased Plagiarism **0**

Word Count 352

Records Found 0

CONTENT CHECKED FOR PLAGIARISM:

Problem Statement:

Creating a website for paralyzed patients which converts the intent of eye blinks to text and speech.

Objective:

To facilitate impaired people with a means to communicate independently:

Many paralyzed patients already use eye blinks as a means of communication which can be usually interpreted and decoded by caretakers and nurses. The ALS association even offers a communication guide that relies on eye blinks. With our project, we intend to automate this task. Our software would read the patient's eye blinks and converts them into text and speech respectively. This would allow patients to express their thoughts and ideas with absolute independence. This kind of independence will provide to be morally uplifting and patients who were earlier reticent to communicate might start communicating more often.

To provide a financially accessible solution:

Various companies are coming up with software solutions to track eye movements and blinks and to interpret and decode them. These software devices are usually powered by expensive hardware. Though these devices undoubtedly facilitate LIS victims to communicate, they can be significantly costly and hence inaccessible to people with limited financial means. We intend to overcome that by developing pocket- friendly equipment. The only required peripheral is a basic

webcam. Hence, this software intends to cater to immobilized patients of all classes.

Scope

The project captures live stream video using webcam of the patient and effectively captures the eye blinks of the patient.

The eye blinks captured are converted to the text which would be displayed at the bottom of the screen. This text is converted to speech to facilitate transferring crucial information to a person who isn't present on the patient's immediate side.

A few assumptions have been made that include, the user's face and eyes being completely present in front of the camera at all times and the user's face being away from the screen at a particular set distance at all times in order to detect and identify eye blinks. These assumptions render the system incompetent to detect eye blinks if the eyes of the user are partially visible or not at a proper set distance.

MATCHED SOURCES:

Report Generated on **April 27, 2021** by prepostseo.com



PLAGIARISM SCAN REPORT

Date April 27, 2021

Exclude URL: NO



Unique Content **93%**

Plagiarized Content **7%**

Paraphrased Plagiarism **0**

Word Count 582

Records Found 4

CONTENT CHECKED FOR PLAGIARISM:

A software requirements specification (SRS) is a description of a software system to be developed. It lays out functional and nonfunctional requirements, and may include a set of use cases that describe user interactions that the software must provide. [7]A software requirements specification (SRS) is a comprehensive description of the intended purpose and environment for software under development.

Functional requirements define the basic system behaviour. Functional requirements are the product features. They generally include the input calculations, data manipulation and processing.

For our system functional requirements are: -

Setting up the keyboard on start

Resetting the column and row intervals after each selection

Detecting blinks accurately

Selecting text accurately

User Requirements

The user shall use a virtual keyboard that shows the rows and columns with animation.

The user will have a view of the selected text.

Nonfunctional requirements specify the quality attributes of a software. They judge the system based on different characteristics.

Reliability - The system should be reliable and true to its content. It should provide accurate mapping of brain signals into proper robotic actions.

Robust - The system should be resistant enough to all sorts of natural as well as physical damage, which in turn leads to less cost and maintenance.

Portable - The device should be small enough to be carried easily anywhere.

Cost - This is the most important function. System should be cost efficient compared to other alternatives available in the market.

Minimalistic Design - The system interface and design must be user friendly to make it more efficient to use.

Usability - System should be easy to use. Minimal training must be provided.

Safety - Since the device is designed for the differently-abled, the system must be free from all parts which are injurious to the body.

Availability - The system should have good connectivity to the internet so that all the interpreted data can be stored in the cloud.

Camera - This is the main component of the system. Without the access to the camera the system won't work. The camera can be a normal web camera attached to a PC or laptop. It can also be the front camera of a smartphone. The camera quality should be above average quality for better understanding.

Anaconda- Anaconda is a free and open software used for performing machine learning programs efficiently. It provides a good package and module management facility which is better for a novice learner.

Flask- Flask is a web application framework written in python. Flask is based on Werkzeug and Jinja2 template engine. This is quite easy to establish a connectivity with the website's front end and back end.

OpenCV- OpenCV (Open Source Computer Vision Library) is a free software for academic and commercial use. It supports various programming languages like C++, Python and Java and is compatible to work on almost all operating systems like Windows, Linux, Mac etc. OpenCV is used by the camera to capture images and convert them into suitable formats.

Dlib- Dlib is a C++ toolkit consisting of different modules suitable for machine learning projects.

Generally, Dlib is used for real time applications.

JavaScript, HTML, CSS3, Bootstrap & Ajax - JavaScript & HTML are programming languages used to build dynamic web applications. Ajax is a tool used to connect to the server without reloading the page. We have used CSS3 and Bootstrap to style the HTML document

Python3 -Python is an interpreted, high-level general-purpose programming language. For our implementation it was used for capturing the eye blinks.

MATCHED SOURCES:

[en.wikipedia.org](https://en.wikipedia.org/wiki/Software_requirements_specification) - 3% *SimilarCompare*

[https://en.wikipedia.org/wiki/Software_requirements_specific....](https://en.wikipedia.org/wiki/Software_requirements_specification)

[reqtest.com](https://reqtest.com/requirements-blog/functional-vs-non-functional-requirements/) - 2% *SimilarCompare*

[https://reqtest.com/requirements-blog/functional-vs-non-func....](https://reqtest.com/requirements-blog/functional-vs-non-functional-requirements/)

[www.chegg.com](https://www.chegg.com/homework-help/questions-and-answers/ma) - 1% *SimilarCompare*

[https://www.chegg.com/homework-help/questions-and-answers/ma....](https://www.chegg.com/homework-help/questions-and-answers/ma)

[steelkiwi.com](https://steelkiwi.com/blog/requirements-why-it-important/) - 1% *SimilarCompare*

<https://steelkiwi.com/blog/requirements-why-it-important/>



PLAGIARISM SCAN REPORT

Date April 27, 2021

Exclude URL: NO



Unique Content **99%**

Plagiarized Content **1%**

Paraphrased Plagiarism **0**

Word Count 319

Records Found 1

CONTENT CHECKED FOR PLAGIARISM:

Architecture Overview

Fig 4.1- Architecture Overview

The vital goal of this project is to take eye blinks of the affected person as input and to transform them to textual content and speech. The same has been depicted by making use of the architecture diagram as shown. The affected person's eye blinks would be examined as input via the Webcam that is then passed onto the image processing module. Then, with the help of the user interaction module, those eye blinks are transformed to textual content after which ultimately to speech.

Detailed Architecture - This provides an overview about how the system communicates with the client and server.

Fig 4.2- Detailed Architecture

Eye Blink Detection Module

The live video captured by the camera is sent to the blink detection module. This processes the video by extracting the facial landmarks using OpenCV, Dlib and other machine learning libraries.

Fig 4.3- Eye Blink Detection Module

User Interaction Module - This provides an overview about the Conda and Flask environment for building the application.

Fig 4.4- User Interaction Module

4.2.5 Text to Speech Conversion Module

Fig 4.4- Text to Speech Conversion Module

Pseudo Codes

Eyeblink Detection Module

Set EAR_CONSEC_FRAMES & EAR_THRESHOLD

While true

Grab The frame Resize and grayscale

Loop across the faces as the come in the frames

Calculate the EAR

Create the visualizations across the face

If $EAR > EAR_THRESHOLD$

Increment Counter

If $Count > EAR_CONSEC_FRAMES$

Write into the file

User Interaction Module

1. Initialize keyboard and screen

2. If blink Detected then Select particular column

3. Set column Interval

4. If blink is detected again then select row element

5. Add selected element to the selected text

Text to Speech Conversion Module

Once the text is printed on the screen, click on the speech button.

If it is a coherent sentence then complete output would be produced as it is.

If it is a mixture of random words then the output would be produced individually.

MATCHED SOURCES:

www.researchgate.net - <1>Compare

<https://www.researchgate.net/figure/RFB-Hierarchical-Archite...>

Report Generated on **April 27, 2021** by prepostseo.com

PLAGIARISM SCAN REPORT

Date April 27, 2021

Exclude URL: NO



Unique Content **89%**
 Plagiarized Content **11%**
 Paraphrased Plagiarism **0**

Word Count 1,135
 Records Found 7

CONTENT CHECKED FOR PLAGIARISM:

This module introduces us to the roadmap of our project. It gives the detailing of tasks undertaken. The project work began in the month of August 2020 and would go on up to April 2021. So far, we have completed the phase 1 of the project which includes Literature Survey, Understanding the pitfalls of the existing technologies and defining our project scope, objective, timeline and the fundamental modules in our project. Phase 2 will include implementation of these modules and deployment of the project. The following tables and Gantt chart would provide an in-depth understanding of the tasks undertaken.

System Design

Systems design is the process of defining elements of a system like modules, architecture, components and their interfaces and data for a system based on the specified requirements. It is the process of defining, developing and designing systems which satisfies the specific needs and requirements of a business or organization. A systemic approach is required for a coherent and well-running system. Bottom-Up or Top-Down approach is required to take into account all related variables of the system. A designer uses the modelling languages to express the information and knowledge in a structure of system that is defined by a consistent set of rules and definitions

The project has been split into 3 major modules for the sake of simplicity while implementation. The following subsections will elaborate on all the pre-requisite libraries downloaded, the module implementations and the additional technologies used.

6.2.1.2 Installation of prerequisite libraries

Numpy - pip install numpy

Sklearn - pip install sklearn

Flask - pip install flask

Pandas - pip install pandas

Matplotlib - pip install matplotlib

Install Opencv and Dlib - conda create --name opencv-env python=3.6

pip install numpy scipy matplotlib scikit-learn jupyter

pip install opencv-contrib-python

pip install dlib

import cv2

cv2.__version__

import dlib

dlib.__version__

6.2.2 Implementation of Eye Blink Detection Module

Information about the module-

We have made use of Dlib, which is a modern C++ toolkit containing machine learning algorithms and equipment for creating complicated software. This Dlib library was then constructed on top of python for our software. Face detection was accomplished with the use of dlib. After that, the eye area was localized from the facial area with the use of Eye Aspect Ratio (EAR). Each eye was represented with the aid of a set of 6(x,y) coordinates, starting at the left corner of the eye and then working around in a clockwise direction. These form 2 sets of vertical landmarks and a set of horizontal landmarks. These landmarks are denoted in the following picture-

The Equation that helps derive the relationship between these coordinates can be given as-

Where, $p_1, \dots, p_6$ are 2D facial landmark locations. In this equation, the numerator computes the distance between vertical landmarks while the denominator computes the distance between horizontal landmarks. The eye aspect ratio is about steady whilst the eye is open however will swiftly fall to 0 while a blink is taking place. As indicated in figure 3, while the eye is completely open, the Eye- Aspect Ratio is huge and relatively consistent over time. Once someone blinks, it decreases dramatically drawing close 0. The bottom figure plots a graph of the eye aspect ratio over time for a video clip. As we can see, the eye element ratio is consistent, then swiftly drops near 0, then will increase again, indicating a single blink has taken place.

Now in order to determine if a blink has taken place, we've got defined variables- threshold consecutive frames. If the eye aspect ratio falls under a certain threshold and then rises above the threshold, then we register a blink—the EYE AR THRESH is this threshold value. Another constant is EYE AR CONSEC FRAME which is set to a selected value which was three for our application indicates that 3 successive frames with an eye aspect ratio much less than EYE AR THRESH must happen in order for a blink to be registered. Another variable specifically COUNTS and TOTAL had been initialised. The TOTAL variable gave the variety of blinks detected whilst this system was running. The COUNT was used to test whether or not the EYE AR CONSEC FRAME constraint was met or not.

Screenshots of Implementation

Info about the module

User Interaction Module The implementation of the user interaction module consisted of the subsequent steps- 1) Initializing the keyboard, 2) Setting interval for the column, 3) Setting interval for row upon column selection 4) Blink detection. We have made use of JavaScript code for the initialization of the keyboard. It consisted of the init() function which specified if the CAPS should be on or not. After the initialization of the keyboard, we set the column interval that highlighted a column in every 2- second delay. If there has been a blink interrupt at some point of this time then that column was selected after which we started the row interval for that specific column. For the selection of the row, selectRow() function was implemented which after detection of the blink selected a particular key and displayed it. In order to fetch detection records from the Blink detection module, we have used AJAX to send requests to the Flask that is linked to the Blink detection module using FIFO. In each 1ms AJAX dispatched a request to look if a blink was detected or not. Once the blink was detected we had to change the front end. We made use of the detection () function that did the essential changes. This function would either write to the display screen the key selected if it was a number or an alphabet or would perform an operation if an action key had been selected. An action key included the Backspace, Caps, and Tab. The detection function was used to clear the intervals and reset the keyboard to its preliminary stage.

Screenshots of Implementation

6.2.4 Implementation of Text to Speech Conversion Module

In order to add a text to speech feature on our web page using JavaScript, we have used a Web Speech API, to synthesize text and convert it into speech. SpeechSynthesisUtterance is the interface in which we actually create the speech or utterance using the text provided, setting a

language type, volume, pitch of the voice, rate of speech etc. The function's name is TexttoAudio. This JavaScript method consists of different properties like language, pitch, rate, volume, text and voice. Once the text is passed in this function, it is read and further passed on to the speech synthesis interface. Then as per how the values have been set, the properties of this method, speech output would be produced. The gender of the speech would depend on the particular system's default voice.

Screenshots of Implementation

6.2.4 Implementation of Website

6.2.6 Linking of Website with User Interface

MATCHED SOURCES:

[economictimes.indiatimes.com](#) - 2% *Similar*Compare

<https://economictimes.indiatimes.com/definition/waterfall-mo....>

[blog.bicom.net](#) - 2% *Similar*Compare

<https://blog.bicom.net/system-design-using-process-flow-to-....>

[askinglot.com](#) - 1% *Similar*Compare

<https://askinglot.com/what-is-system-design-process>

[quizlet.com](#) - 1% *Similar*Compare

<https://quizlet.com/26744336/science-chapter-3-flash-cards/>

www.slideshare.net - *<1>Compare*

<https://www.slideshare.net/VinitaMohan1/how-to-install-numpy....>

www.ijert.org - *<1>Compare*

<https://www.ijert.org/eye-blink-detection-method-for-disable....>

www.tutorialspoint.com - *<1>Compare*

<https://www.tutorialspoint.com/how-to-install-matplotlib-in-....>

Report Generated on **April 27, 2021** by prepostseo.com



PLAGIARISM SCAN REPORT

Date May 05, 2021

Exclude URL: NO



Unique Content **97%**

Plagiarized Content **3%**

Paraphrased Plagiarism **0**

Word Count 1,445

Records Found 5

CONTENT CHECKED FOR PLAGIARISM:

In this module, the facial region was isolated by making use of Haar Cascade Classifier. Landmark detection was performed upon which the eye region was located. The eye region was further identified by calculating the eye aspect ratio and the eye blinks were noted. The input given to this module was the face of the user which was taken as the input via the webcam. The result obtained was the eye blinks occurring which were successfully noted and the number of eye blinks taking place were noted in python file

In this module, the eye blinks that have been identified in the previous module are used to navigate across the keyboard which is present on the screen. The navigation takes places across the keyboard with the navigator moving column-wise first and then row-wise after a particular column has been selected. The input given to this module was the eye blinks that were being captured via the live stream video. The result obtained was the text which had been typed with the help of the blinks.

In this module, the main aim was to convert the text that the user has formulated with the help of his eye blinks by navigating across the keyboard to speech. The input given to this module was the text that the user had typed with the help of blinks. The result obtained was the text that was converted to speech.

Eye blink detection testing has been done to check the correctness of the eye blink module which was implemented. The main goal of eye blink detection module was to successfully isolate the eye part from the facial region and capture the eye blinks of the users. The following table shows the various test cases for eye blink detection module. Eye blink detection is used to eventually select

alphabets for the keyboard. The number of milliseconds that the number of blinks was done and should generate a blink signal is 1 second and was set as the threshold. We have also checked if the system is able to distinguish between different kinds and colours of eyes. The following table shows the various test cases for eye blink detection module-

Sr. No

Test Case Description

Input

Expected Output

Actual Output

Result

1.

Eyes Shut

10 pairs of eyes at different instances.

No blinks should be detected

Blinks were not detected

PASS

2.

Eye Detection

Brown Eyes

Blinks should be detected

Blinks were detected

PASS

3.

Eye Detection

Black Eyes

Blinks should be detected

Blinks were detected

PASS

4.

Eye Detection

Grey Eyes

Blinks should be detected

Blinks were detected

PASS

5.

Eye Detection

Eyes containing cataract

Blinks should be detected

Blinks were not detected

FAIL

6.

Eye Detection

Cataract operated eyes

Blinks should be detected

Blinks were detected

PASS

User Interaction Module testing had been implemented to check the correctness of the functioning of the module which was implemented. The main aim of this module was to read the eye blinks as input and to use them for navigating across the keyboards. Parameters were defined for this navigation. Initially, a blinker started its motion on the keyboard from left to right in a cyclic pattern alongside the columns present on the keyboard. Upon, blinking of the eyes once, a particular column on the keyboard was selected and then the blinker began its navigation in the rows of that particular column. The accuracy was then determined by the number of words correctly typed. The following table illustrates the functioning of this module-

Sr.

Initialization of the keyboard.

The clicking of the launch button on the website.

The keyboard should have been initialized.

The keyboard was initialized.

PASS

2.

Navigation of the blinker across columns.

Initialization of keyboard.

The blinker should have navigated across columns.

The blinker navigated across columns.

PASS

3.

Navigation of the blinker across rows.

Selecting a particular column.

Navigation of rows upon column selection.

Blinker navigated across the rows on column selection.

PASS

4.

Typing of alphabets using eye blinks

The eye blinks of the user.

Selection of alphabets on the keyboard

The alphabets were correctly selected with the help of blinks

PASS

5.

Typing of numbers using eye blinks

The eye blinks of the user

Selection of numbers on the keyboard

The numbers were correctly selected with the help of blinks

PASS

6.

Typing of special keys using eye blinks

The eye blinks of the user

Selection of special keys on the keyboard

The special keys were correctly selected with the help of blinks

PASS

Text to Speech Conversion Module testing had been implemented to check the correctness of the functioning of the module which was implemented. The main objective of this module was to correctly translate to speech, the text that was given as input. The following table illustrates the different test cases tried for the same-

Sr.

Functioning of the speak button

Clicking on the speak button.

Translation of the text to speech

The text was converted to speech on clicking of speak button

PASS

2.

Conversion of the alphabet to speech

Clicking on the speak button.

Alphabet to be converted to Speech.

The speech was obtained from the input alphabet.

PASS

3.

Conversion of numbers to speech

Clicking on the speak button.

Number to be converted to Speech

The speech was obtained from the input number.

PASS

4.

Conversion of words to speech

Clicking on the speak button.

Words to be converted to Speech

The speech was obtained from the input words.

PASS

5.

Conversion of random string (consisting of alphabets and numbers) as a coherent word in speech format.

Clicking on the speak button.

Speech to be obtained as coherent words.

The the speech wasn't read coherently.

FAIL

6.

Conversion of random string (consisting of alphabets and numbers) as a randomly read string in speech format.

Clicking on the speak button.

String to be converted to Speech in non-coherent form.

The speech was obtained of in the form of a random string.

PASS

7.

Conversion of sentences to speech

Clicking on the speak button.

Sentences to be converted to Speech

The speech was obtained of the input sentences

PASS

In this testing, all the above-mentioned modules which have now been integrated together were tested as one whole system. So, the main job of this whole system was to take eye blinks as input from the user which were then used to navigate on the keyboard as a part of the user interaction module, with which words were formed which were shown on the output screen and this was eventually converted to The following table shows the working of the same-

Sr.

Intended Text

Aditi Godbole

Aditi Godbole

Aditi Godbole

PASS

2.

Intended Text

Deepika Rao

Deepika Rao

Deepika Rao

PASS

3.

Intended Text

Komal Pawoolkar

Komal Pawoolkar

Komal Pawoolkar

PASS

4.

Intended Text

Comp Eng

Comp Eng

Comp Eng

PASS

5.

Intended Text

NU768

NU768

NU768

PASS

6.

Intended Text

qazwsx123

qazwsx123

qazwsx123

PASS

Live stream videos were taken as inputs during the testing of this module. The high-speed webcam was used to read the video as input from the user. A total of 10 people participated during the trials of the system. The output was reviewed by three individuals. During the review, the individuals watched the participants from aside and noted each time during which the participants blinked and whether the system registered it as a blink or a miss. Every time, the participant blinked and the system correctly identified it, it was classified as a blink and for the occasions on which the system failed to identify the blink, it was noted as a miss. If the system detected a blink and if the participant had not performed the activity of blinking, it was classified as a False Positive.

Now, humans perform blinks almost involuntarily in many instances. Hence, in order to distinguish the spurious blink from an actual one, we have defined a threshold value of 3 seconds. Another variable of consecutive frames was also defined. If a blink did take place for this duration, for a consecutive number of frames, only then it was actually classified as an actual blink and the rest were ignored as spurious blinks. A total of 1000 blinks were noted for testing purpose. Only a total of 5% blinks out of 1000 blinks i.e. 50 blinks were missed and a mere 1% of blinks i.e 10 blinks

were falsely detected. Overall, the eye blink detection module performed with an accuracy of 94%.

MATCHED SOURCES:

www.coursehero.com - <1>Compare

<https://www.coursehero.com/file/p543k78r/Test-case-no-2-Test....>

support.microsoft.com - <1>Compare

<https://support.microsoft.com/en-us/windows/dictate-text-usi...>

github.com - <1>Compare

<https://github.com/IEEE-NITK/Neural-Voice-Cloning>

wicg.github.io - <1>Compare

<https://wicg.github.io/keyboard-lock/>

bugzilla.kernel.org - <1>Compare

https://bugzilla.kernel.org/show_bug.cgi?id=194557



PLAGIARISM SCAN REPORT

Date May 05, 2021

Exclude URL: NO



Unique Content **98%**

Plagiarized Content **2%**

Paraphrased Plagiarism **0**

Word Count 409

Records Found 1

CONTENT CHECKED FOR PLAGIARISM:

As a part of this project, we had proposed a system which would take eye blinks of the patients as an input and would effectively transform them to text and speech. In order to perform the same, we had divided this project into 3 modules namely Eye Blink detection Module, User Interaction Module and Text to Speech Module. As a part of the Eye Blink Detection module, we have captured the eye blinks of the patient and the same was done by the use of Haar Cascade Classifiers and calculation of Eye Aspect Ratio. Based on the threshold limits and consecutive frames a blink was registered. In the User Interaction Module, we have created a keyboard and an output screen. The navigation across the keyboard was performed by a blinker column wise on first blink and row wise on second blink. In Text to Speech module a function TexttoAudio was defined in JavaScript to convert the obtained text from user interaction module into speech. Various parameters like rate, pitch, volume etc. could be adjusted as a part of this module.

The results for each of the module are summarized below-

Eye Blink Module - Eye blinks occurring were successfully recorded and noted. A total 1000 blinks were noted for testing purpose. 5% blinks out of 1000 i.e., 50 blinks were missed and 1% i.e 10 blinks were falsely detected. Accuracy achieved in this module is 94%.

User Interaction Module - The text typed with the help of the blinks is the output. The accuracy was analysed on the basis of navigation of blinker across rows and columns and the words being correctly rendered.

Text to Speech Conversion Module - The text was converted to speech in this module. Accuracy was gauged on the basis of words correctly translated to speech.

Website - Our website is hosted using Flask and Anaconda. HTML and CSS are used to design the frontend. The Eyeblick interface designed using JavaScript has been integrated and is working successfully.

We have performed test cases for all the three modules for various scenarios out of which one case failed in eye blink detection module and one case failed in text to speech module. This led us to discover the shortcomings of the project and hence come up with potential areas which can be improved upon in future. Thus we have successfully curated a system which is cost efficient and is a very effective tool for the user.

MATCHED SOURCES:

www.guru99.com - 2% SimilarCompare

<https://www.guru99.com/front-end-vs-back-end-developers.html>

Publication by Candidates

We have submitted the paper to the below conference and we have received acceptance for the same:

- **International Conference on Software Engineering and Computer Science (ICSECS). ICSECS which would be held on 16th May 2021 in Bengaluru, India.**

Acknowledgement

We are very thankful to our college, New Horizon Institute of Technology and Management for presenting us with this wonderful opportunity of working on this exciting project. The implementation of this project would not have been possible without the guidance, assistance and suggestions of many individuals. We would like to express our deep sense of gratitude and indebtedness to each and every one who has helped us out throughout the course of the project & thus ensuring that this project would be a success.

We heartily thank our Principal, Dr. Prashant Deshmukh, New Horizon Institute of Technology and Management for his constant encouragement and inspiration in taking up this project.

We heartily thank the Head of the Department, Dr. Sanjay Sharma, Dept. of Computer Engineering, New Horizon Institute of Technology and Management for providing us with all the resources and constantly guiding us through the process.

Once again, we would like to heartily thank Dr. Sanjay Sharma, who has also been our project guide for always inspiring us to push our boundaries and explore various new technologies. We would like to say thank you to sir for his constant follow-up and advice throughout the course of the Project work.

We would also like to take opportunity to thank our project coordinator, Ms. Aarathi Kamble, for helping us navigate through difficult parts of formation of the synopsis and for so efficiently organizing and ensuring the smooth running of all the project presentations.

Special thanks to all the staff members of Computer Science Department for their help and kind co-operation.

Lastly, we would thank our parents and friends for their encouragement and support given to us in order to finish this precious work.

EYECON THE SPEAKER FOR PARALYZED

Project by: Aditi Godbole, Deepika Rao, Komal Pawoolkar

Project Guide: Dr Sanjay Sharma

Group No.: C0172103 | Class of 2021 | Department of Computer Engineering



ABSTRACT

- ❖ The fundamental goal of this project is to allow **impaired** and **handicapped** patients to convert their intents which will be gauged using eye blinks and converted to **text** and **speech**.
- ❖ Our project aims to detect the eye blinks of the patient using webcams or mobile front cameras and convert them to text and speech respectively.
- ❖ The interface detects the voluntary eyeblinks and interprets them as commands which are used to select text for the user.
- ❖ This will be simultaneously converted to speech.

INTRODUCTION

- ❖ Paralyzed people cannot control muscle function in one or more muscle groups.
- ❖ In order to facilitate a means of expression for LIS patients, we propose a robust system which will facilitate the patients to convert their expression to text and speech.
- ❖ This project primarily focuses to provide a means of communication to people of all classes without having to worry about financial aspects or cumbersome hardware.

OBJECTIVES

- ❖ To facilitate impaired people with a means to communicate independently.
- ❖ To provide a financially accessible solution

PROBLEM STATEMENT

To convert the intent of eye blinks into text and speech.

ARCHITECTURE



IMPLEMENTATION OF EYE BLINK DETECTION MODULE



- ❖ In this module, the facial region was isolated by making use of **Haar Cascade Classifier**. Landmark detection was performed upon which the eye region was located.
- ❖ The eye region was further identified by calculating the eye aspect ratio and the eye blinks were noted.

IMPLEMENTATION OF USER INTERACTION MODULE



- ❖ In this module, the eye blinks that have been identified in the previous module are used to navigate across the **keyboard** which is present on the screen.
- ❖ The navigation takes place across the keyboard with the navigator moving **column-wise** first and then **row-wise** after a particular column has been selected.

IMPLEMENTATION OF TEXT TO SPEECH MODULE



- ❖ In this module, the main aim is to convert the **text** that the user has formulated with the help of his **eye blinks** by navigating across the keyboard to **speech**.

WEB APP



FUTURE SCOPE

- ❖ The future scope for this project could be, an input can be read accurately even if the face is partially visible.
- ❖ In future, attempts could be made to identify the eye region even in dim or gloomy lighting.
- ❖ Efforts can be made to include frequently used words like please, thank you, medicines, water etc.
- ❖ Additional work can be carried out to include auto suggest options in the keyboard.
- ❖ In the future, attempts can be made to provide support for other regional and global languages as well.

CONCLUSION

- ❖ Thus with the help of this project, we have tried to come up with a solution which is cost efficient and easy to use for the patients so that they feel included in the conversations with the society.
- ❖ We have made use of various technologies like python, flask, ajax, css, html etc. in the implementation of this program.
- ❖ The project like every other technology has its shortcomings which can be improvised upon by working upon the suggestions suggested in the future scope.
- ❖ In conclusion, we believe that our project will provide the patients with an effective tool of communication.

REFERENCES

- ❖ KARKI, M. A., TIMILSINA, M. A., REDDY, M. V. I. K., HEGDE, M. S., & GUIDE, M. (2018). CONVERTING INTENT TO TEXT USING REAL-TIME EYEBLINK DETECTION.
- ❖ Namratha, K. S., Jeevitha, B. S., Sneha, P. H., & Veda, D. Eye blink to voice format communication for paralyzed patients.
- ❖ Goyal, K., Agarwal, K., & Kumar, R. (2017, April). Face detection and tracking: Using OpenCV. In 2017 International conference of Electronics, Communication and Aerospace Technology (ICECA) (Vol. 1, pp. 474-478). IEEE.